

# BFS Traversal

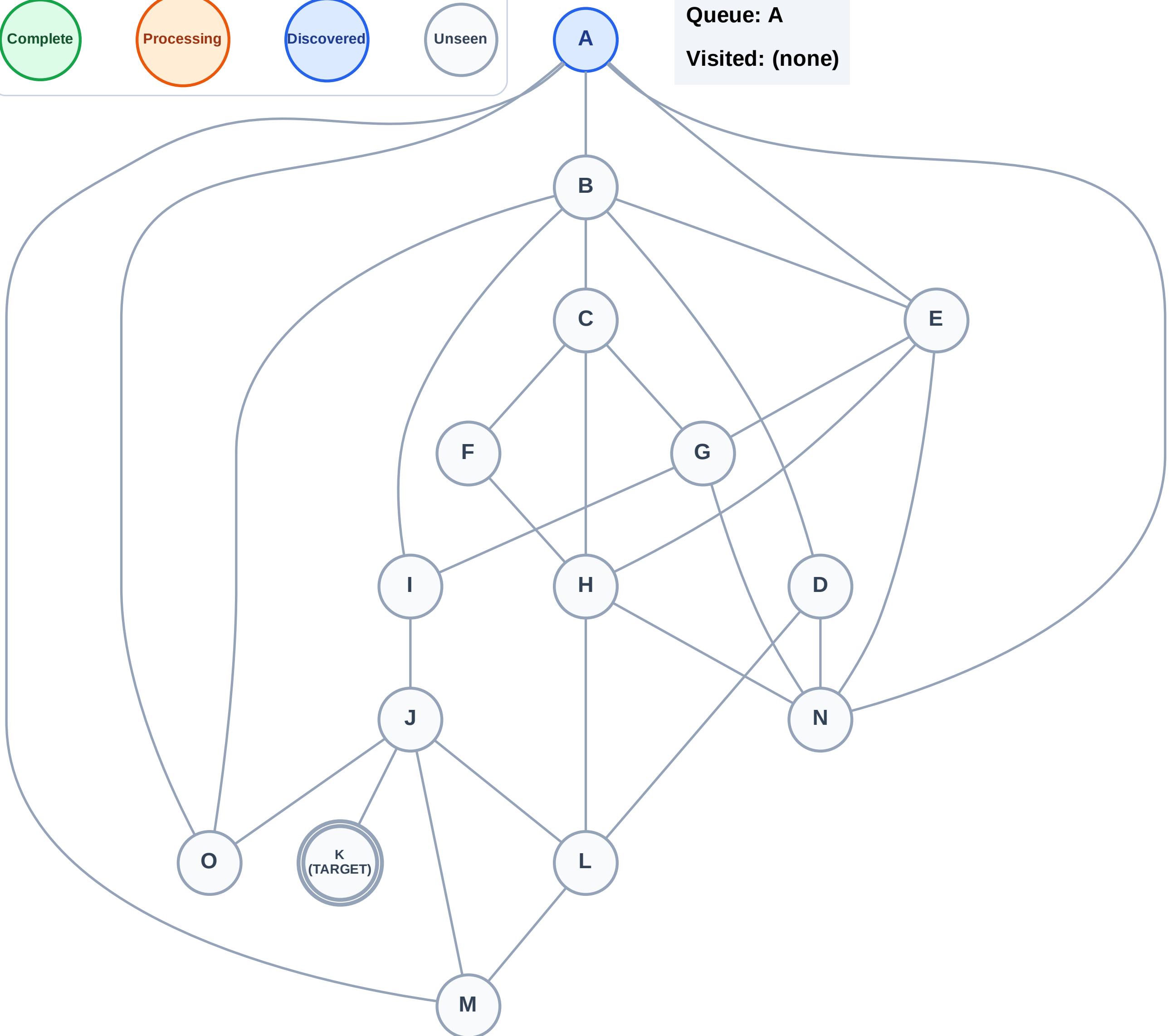
Step 1: Start at A (Seeking target node K)

## Legend



Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

Legend

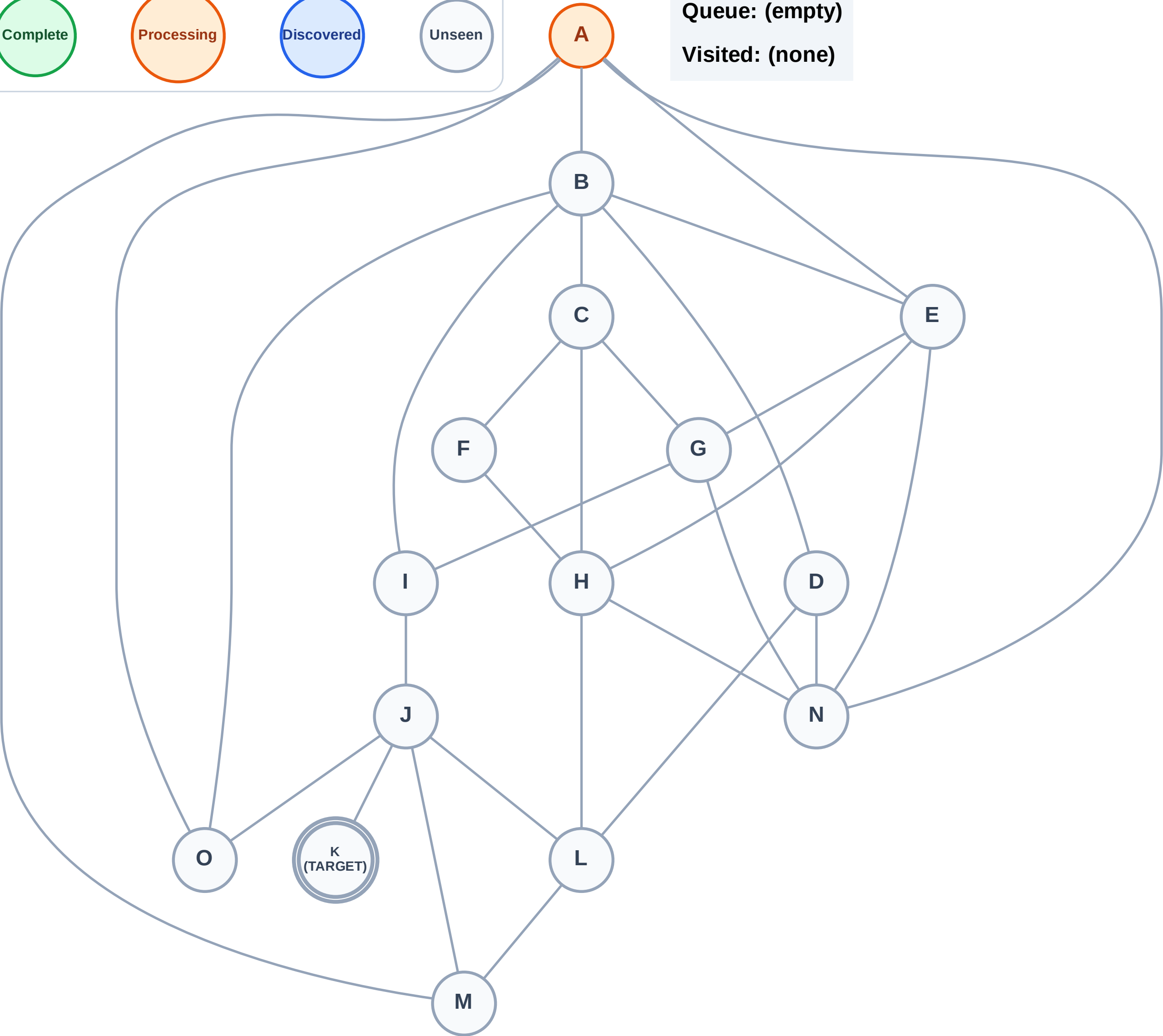
Complete

Processing

Discovered

Unseen

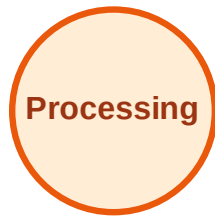
Queue: (empty)  
Visited: (none)



# BFS Traversal

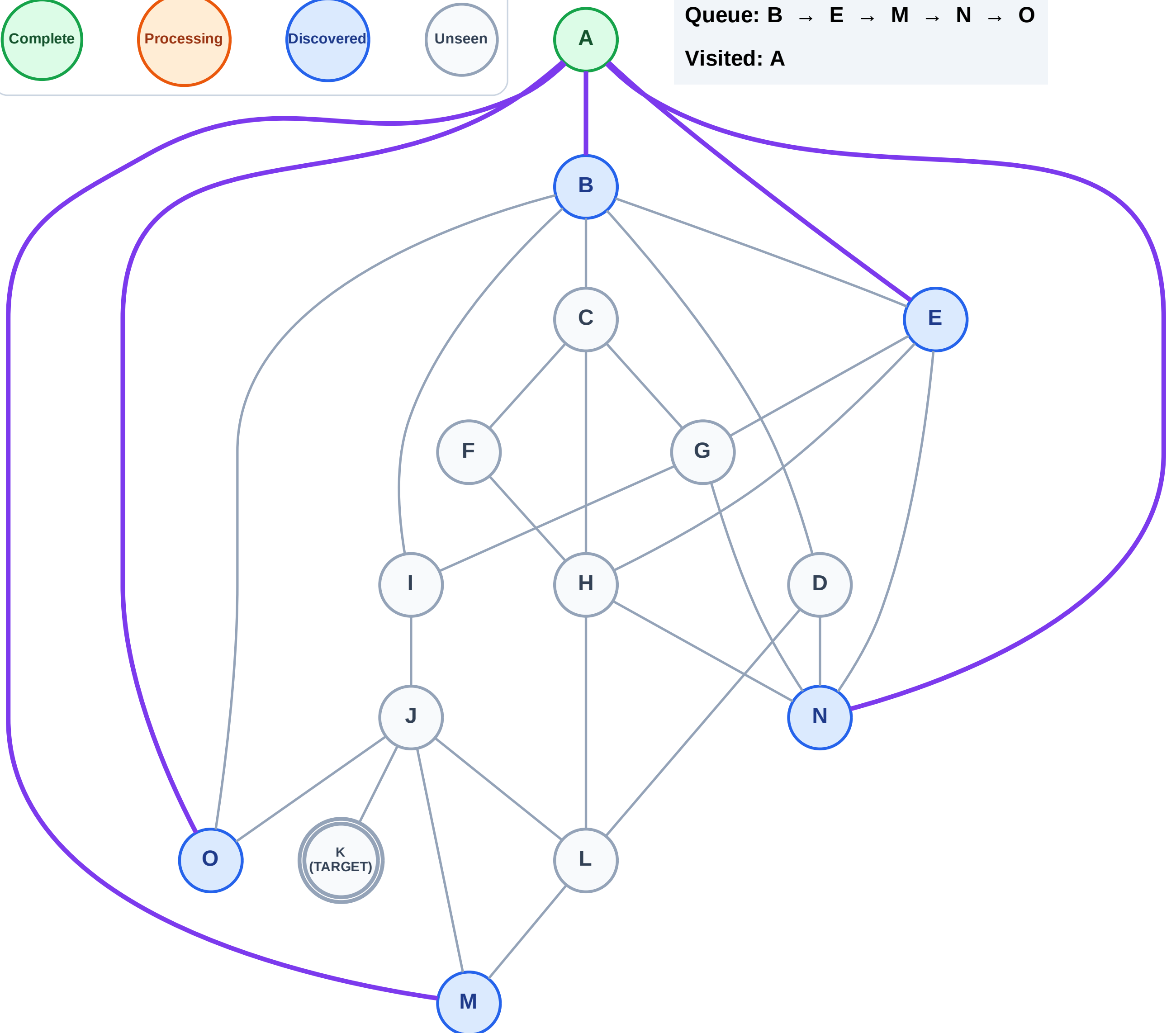
Step 3: Discover B, E, M, N, O from A

## Legend



Queue: B → E → M → N → O

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

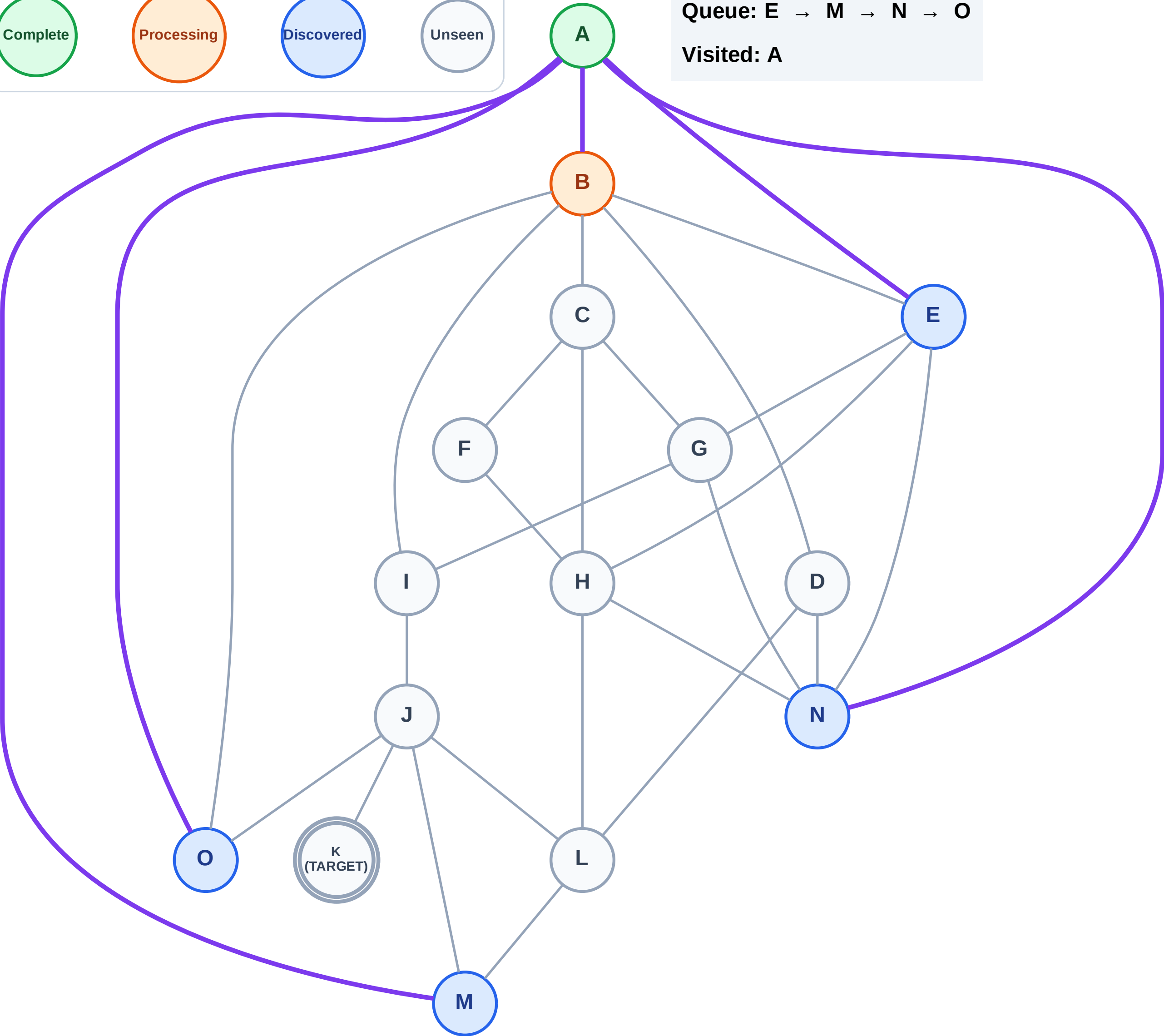
Processing

Discovered

Unseen

Queue: E → M → N → O

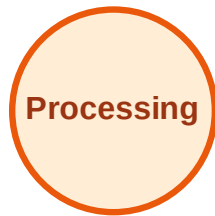
Visited: A



# BFS Traversal

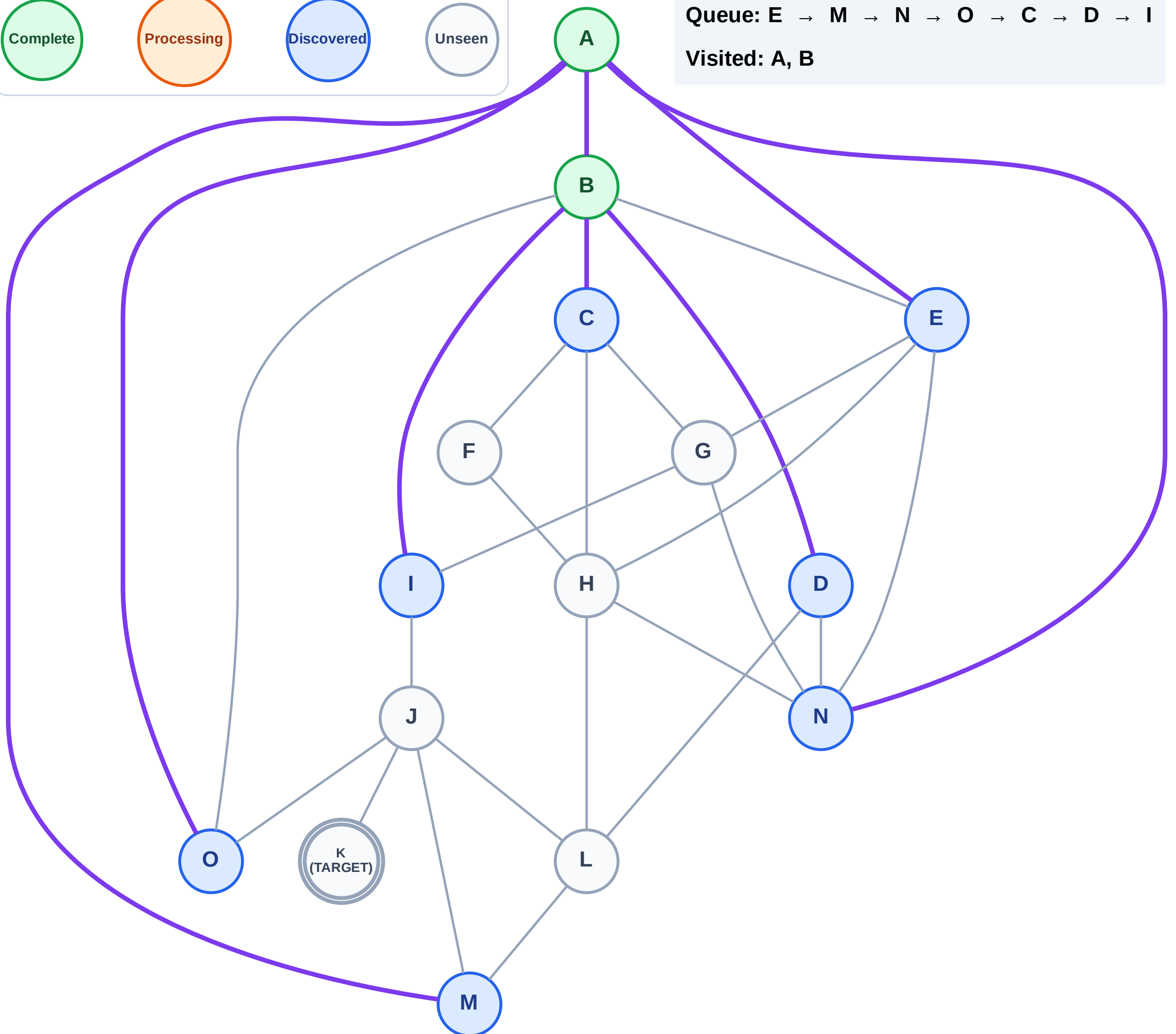
Step 5: Discover C, D, I from B

## Legend



Queue: E → M → N → O → C → D → I

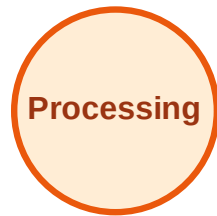
Visited: A, B



# BFS Traversal

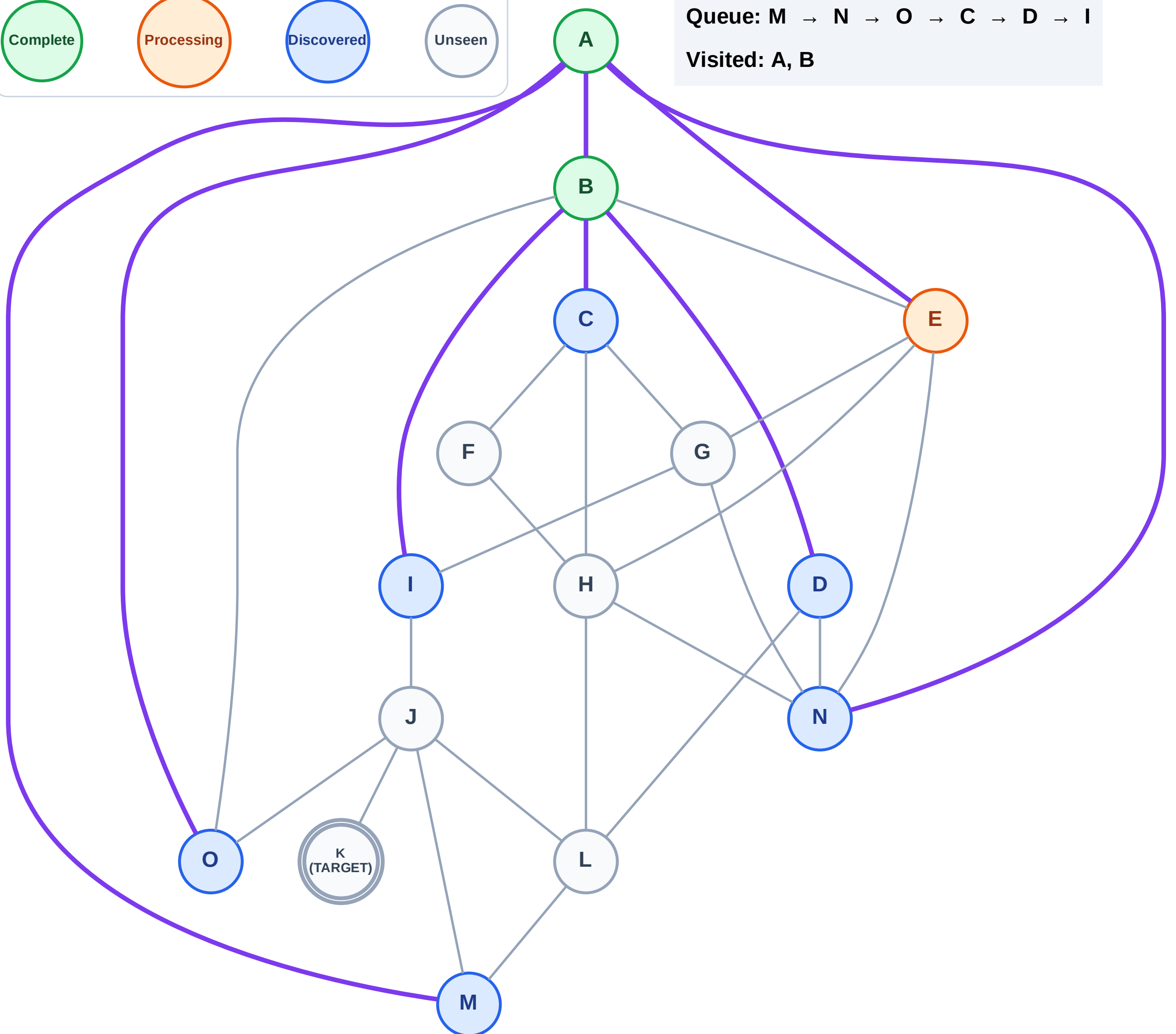
Step 6: Process node E

## Legend



Queue: M → N → O → C → D → I

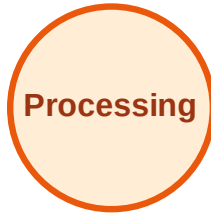
Visited: A, B



# BFS Traversal

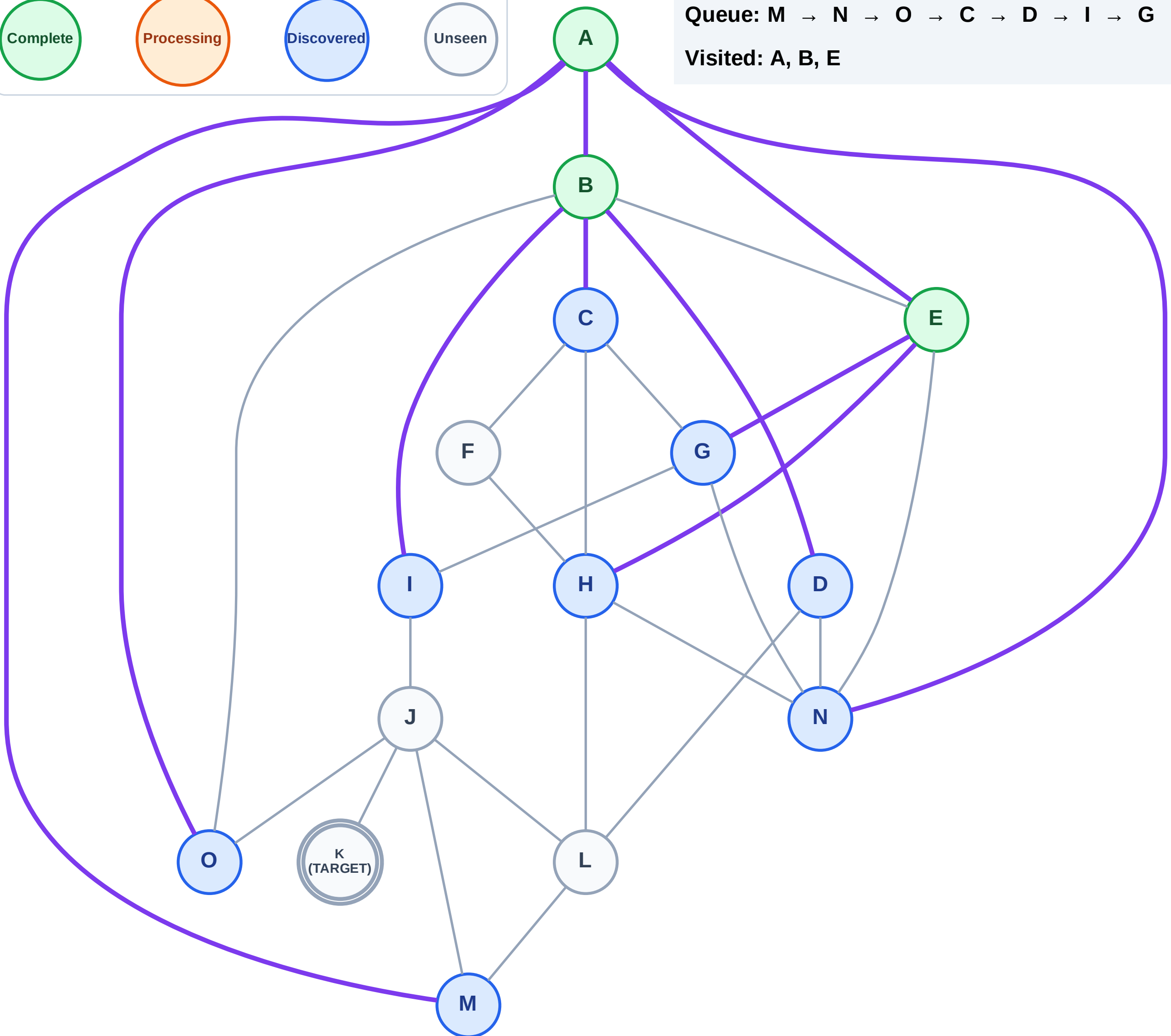
Step 7: Discover G, H from E

## Legend



Queue: M → N → O → C → D → I → G → H

Visited: A, B, E



# BFS Traversal

Step 8: Process node M

Legend

Complete

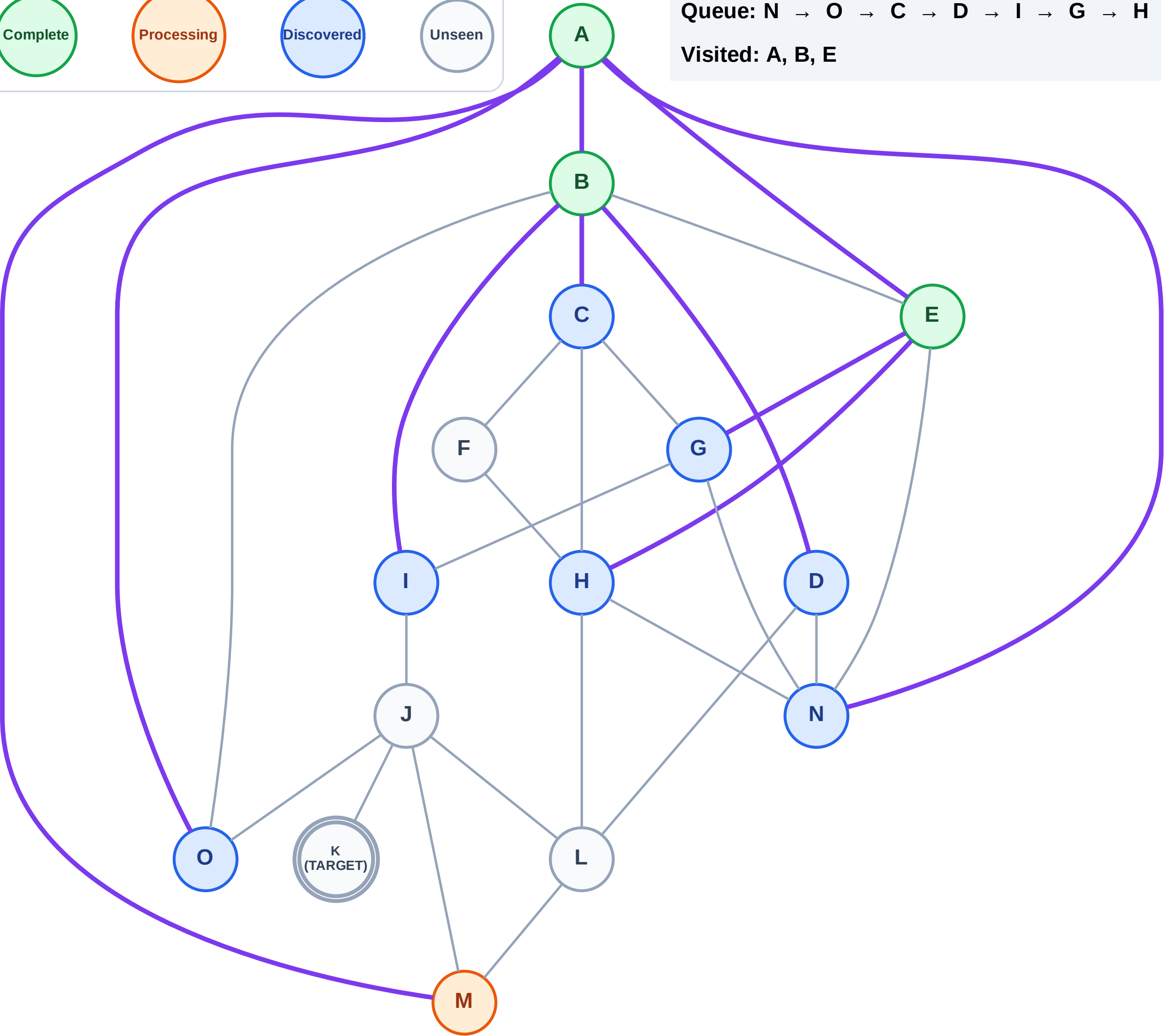
Processing

Discovered

Unseen

Queue: N → O → C → D → I → G → H

Visited: A, B, E



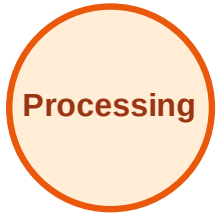
# BFS Traversal

Step 9: Discover J, L from M

## Legend



Complete



Processing



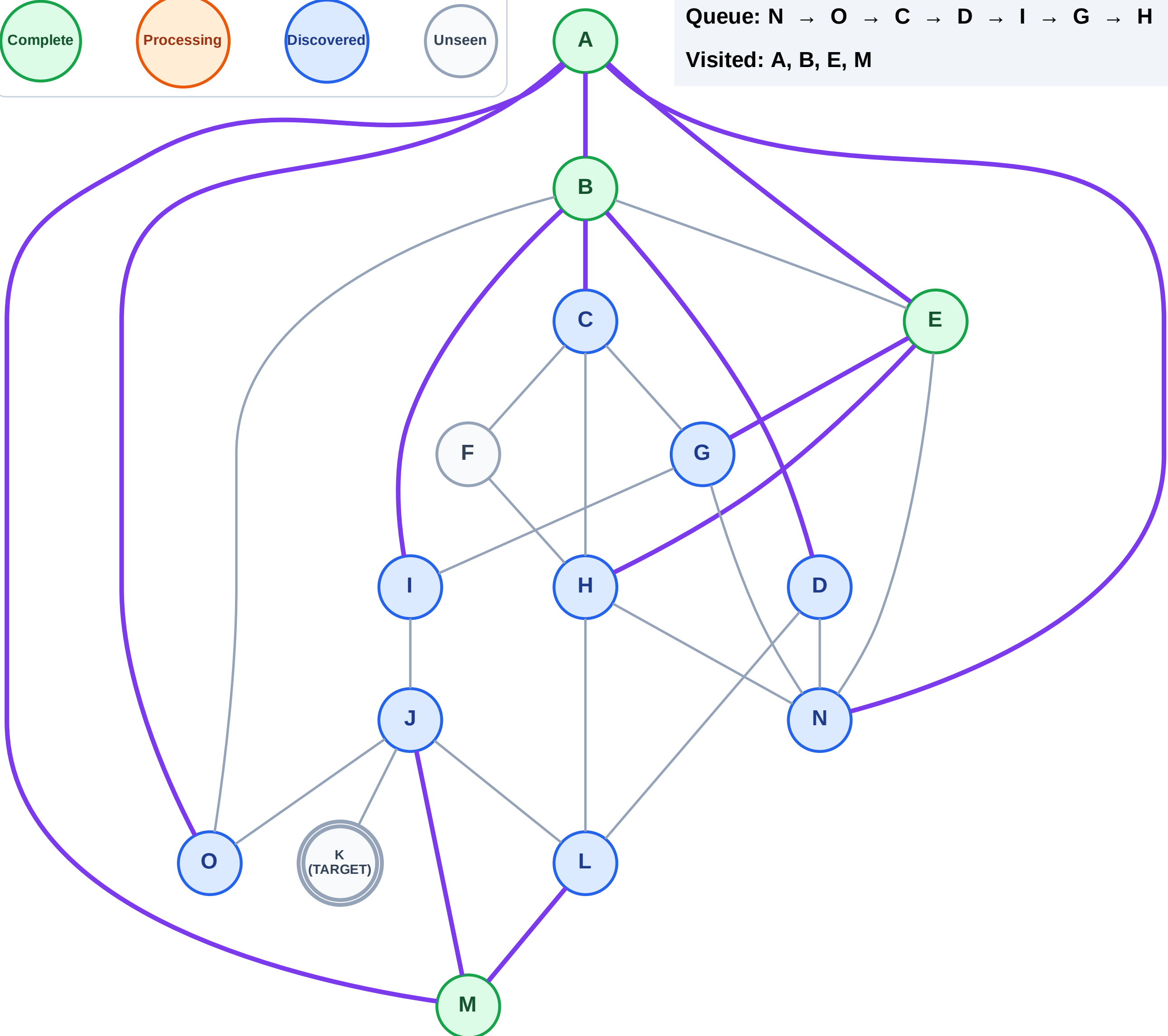
Discovered



Unseen

Queue: N → O → C → D → I → G → H → J → L

Visited: A, B, E, M



# BFS Traversal

Step 10: Process node N

Legend

Complete

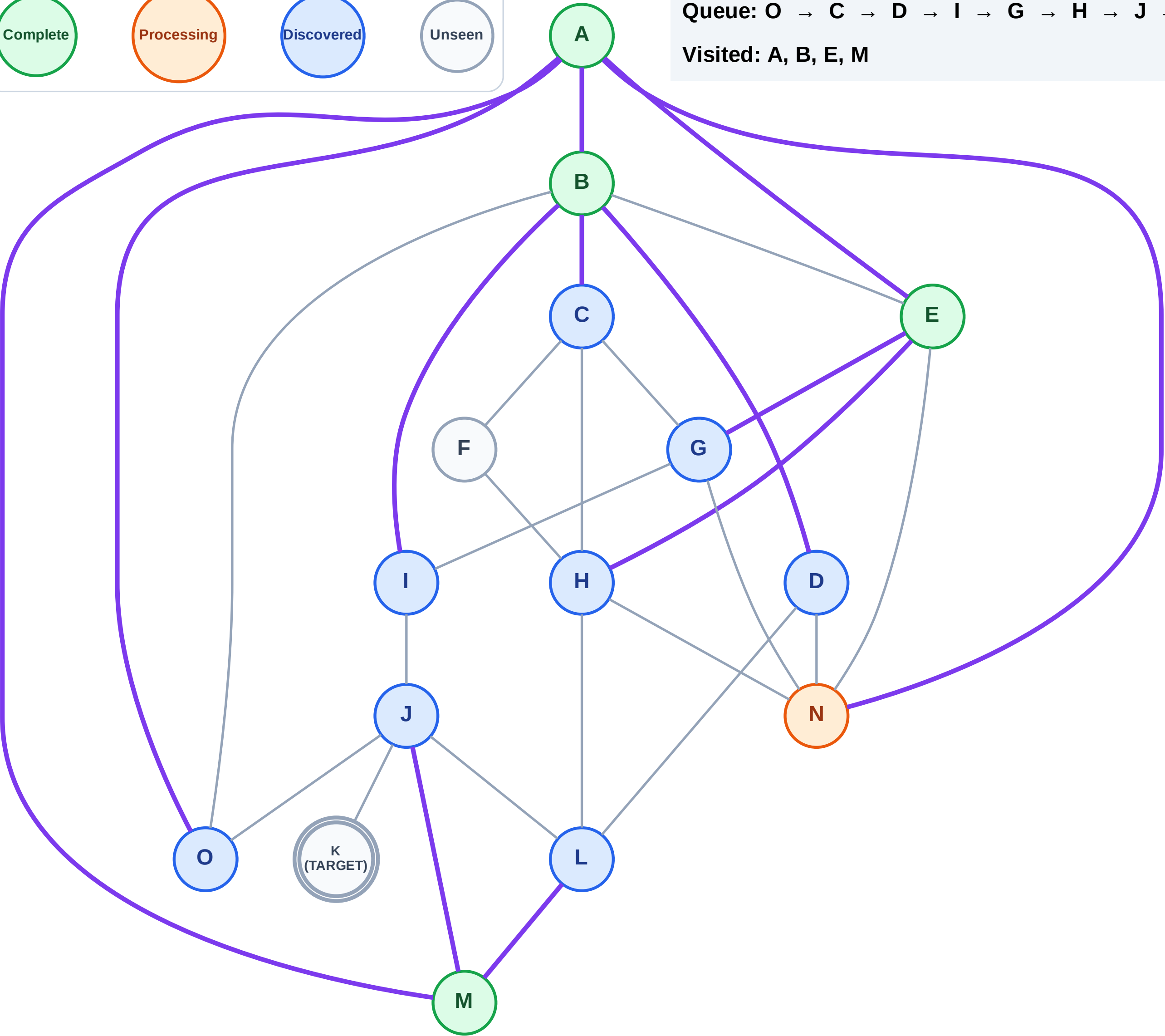
Processing

Discovered

Unseen

Queue: O → C → D → I → G → H → J → L

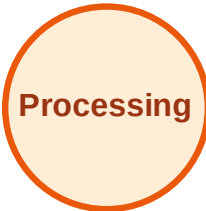
Visited: A, B, E, M



# BFS Traversal

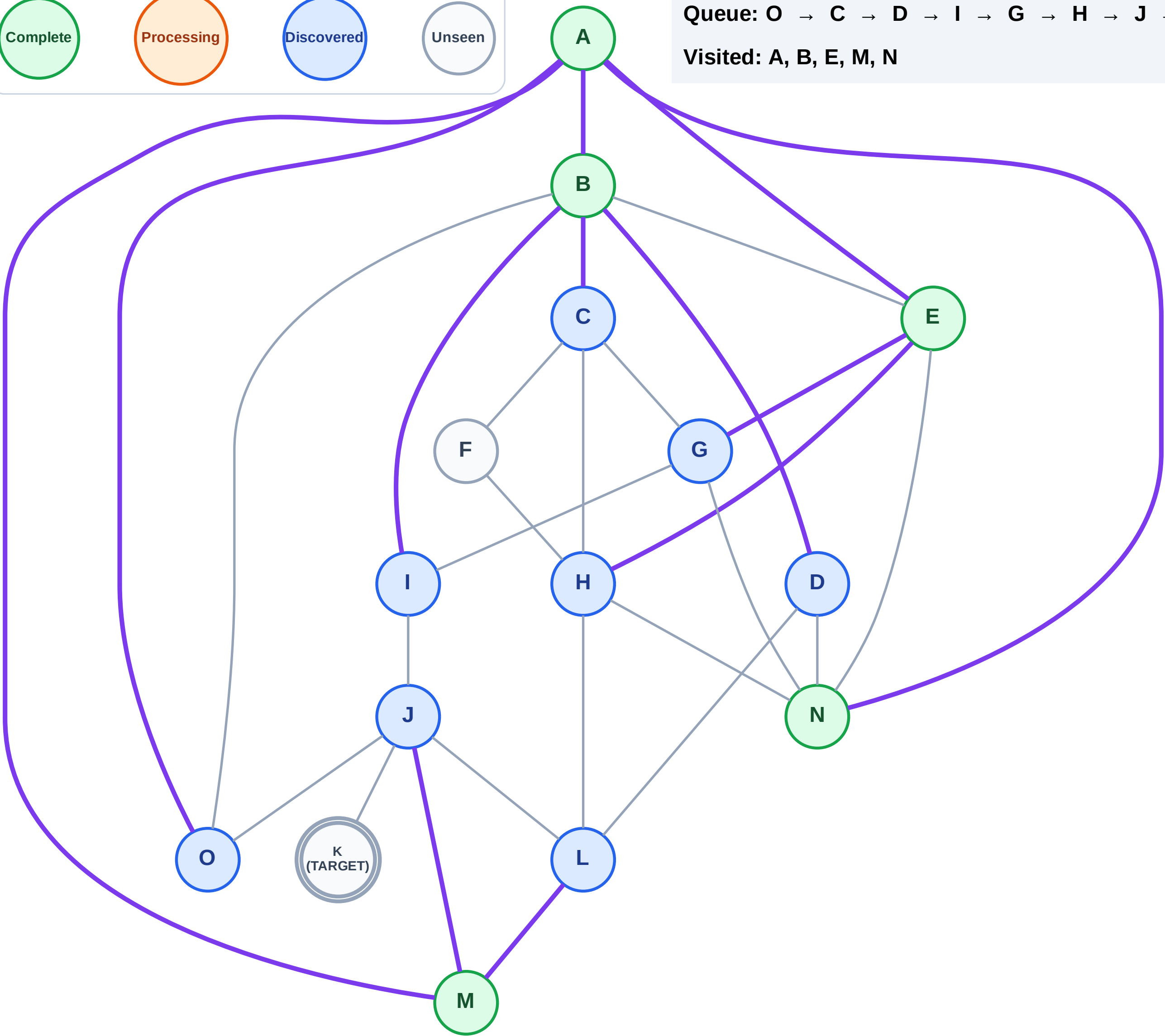
Step 11: No new neighbors from N

## Legend

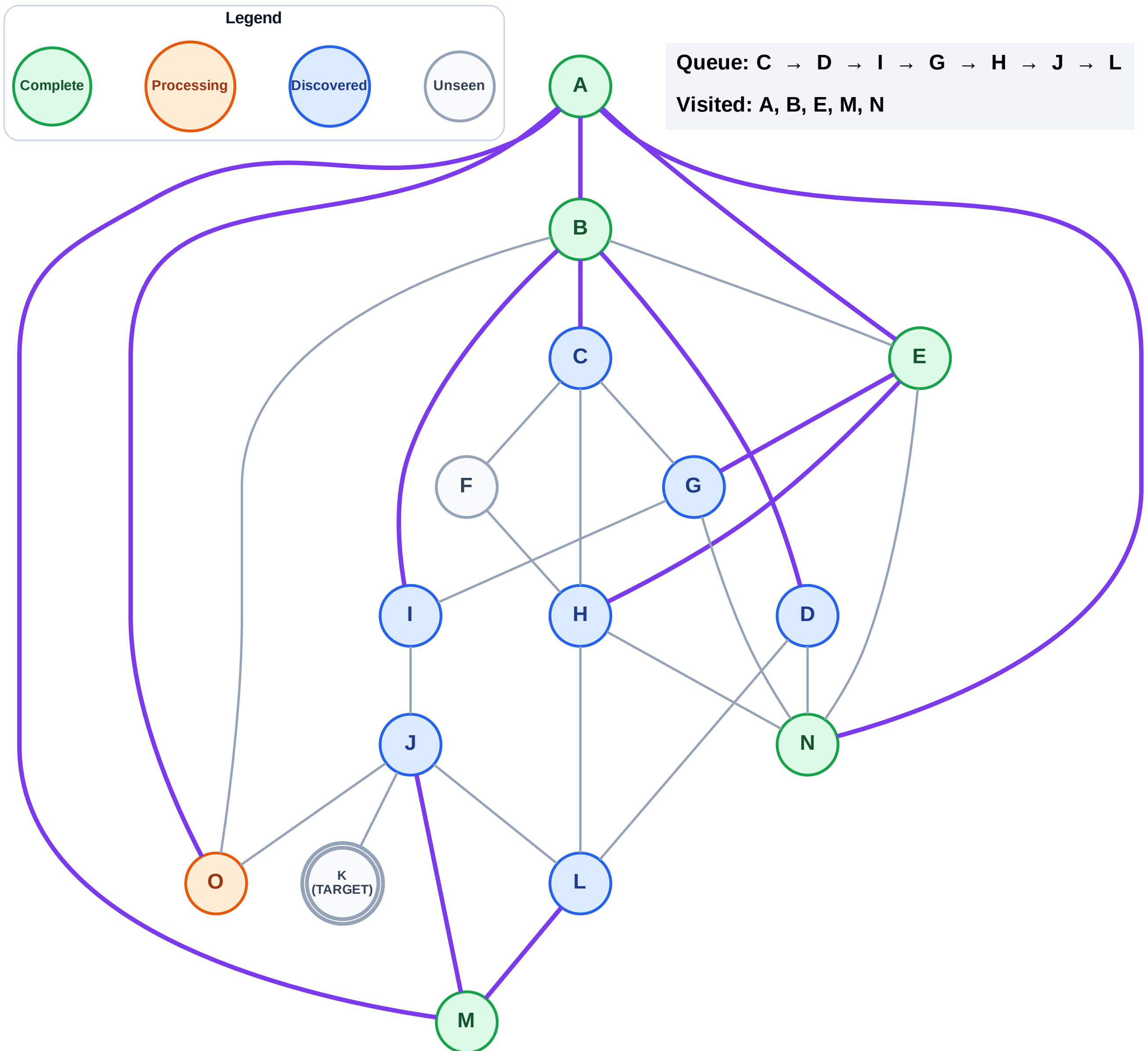


Queue: O → C → D → I → G → H → J → L

Visited: A, B, E, M, N



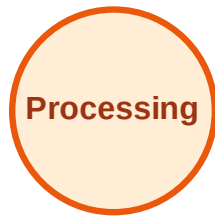
## Step 12: Process node O



# BFS Traversal

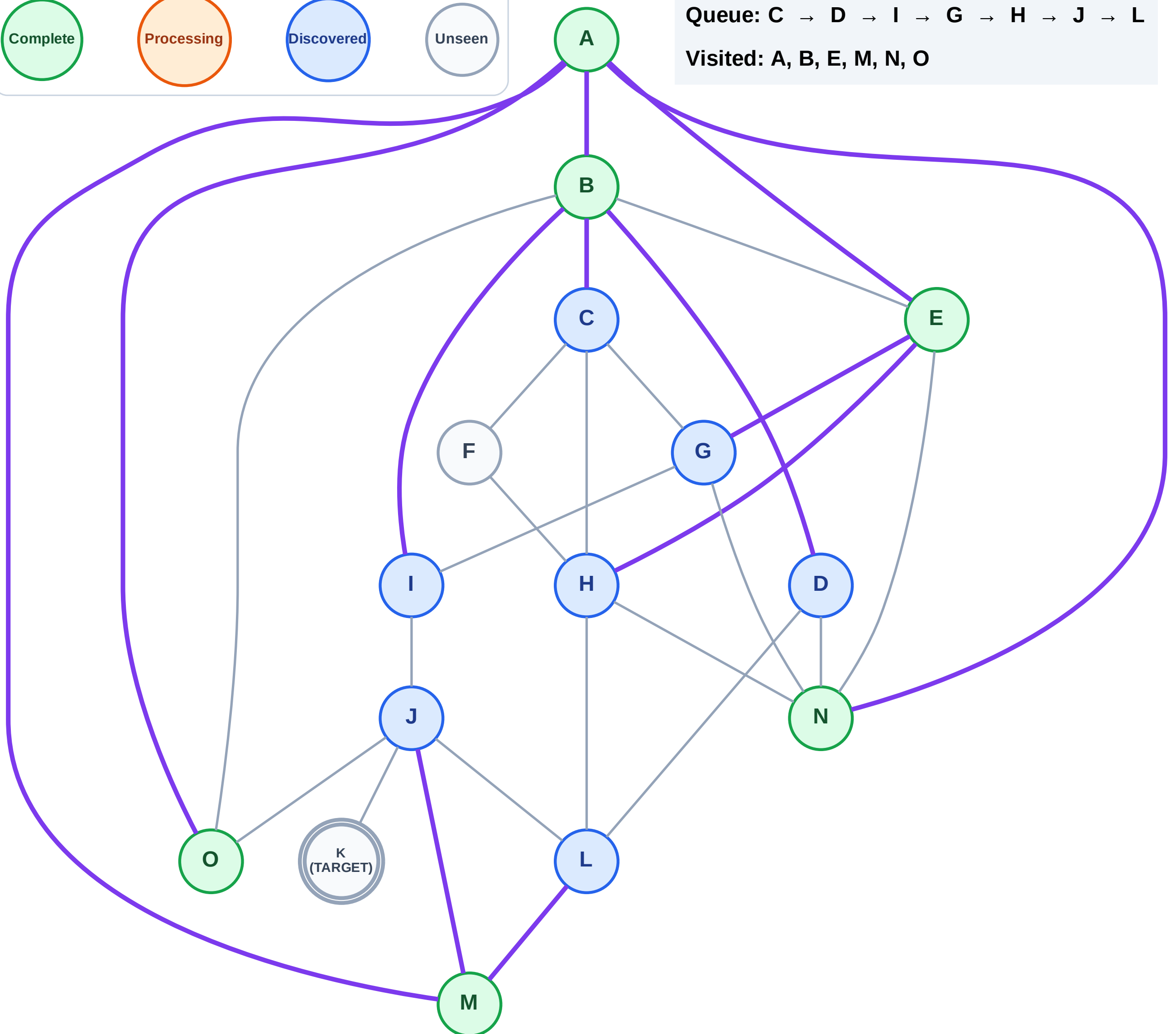
Step 13: No new neighbors from O

## Legend



Queue: C → D → I → G → H → J → L

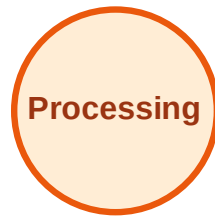
Visited: A, B, E, M, N, O



# BFS Traversal

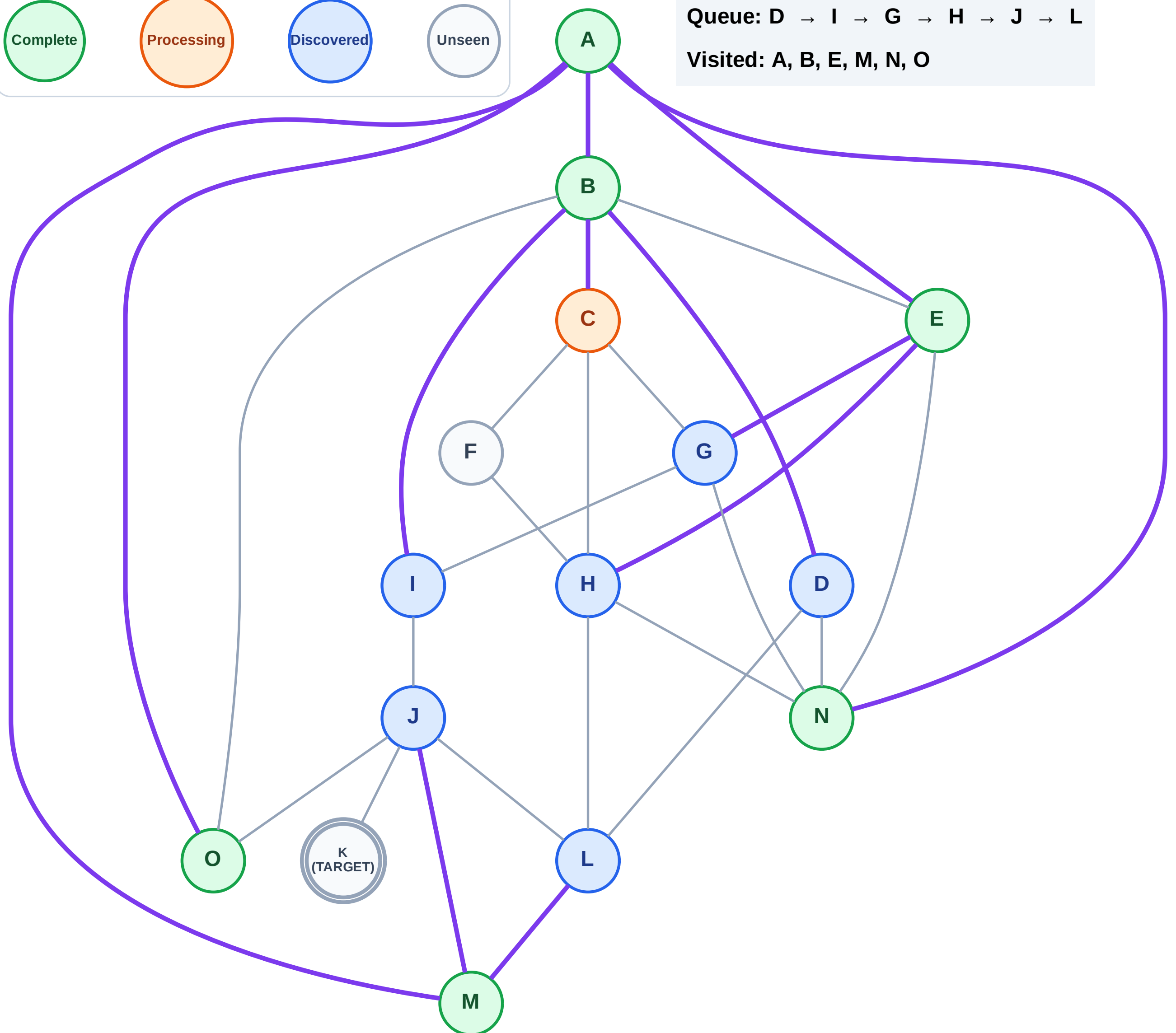
Step 14: Process node C

## Legend



Queue: D → I → G → H → J → L

Visited: A, B, E, M, N, O

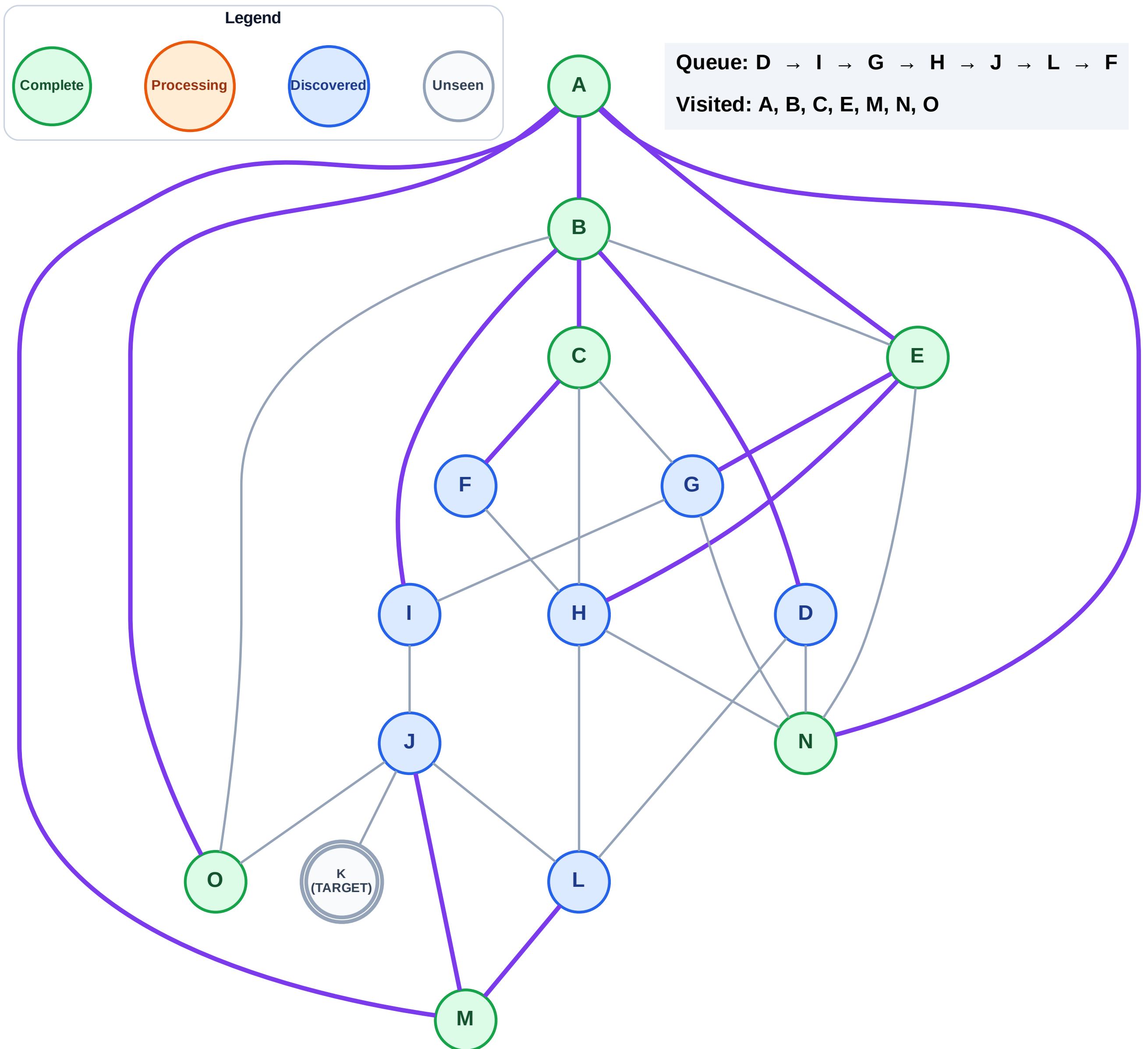


### Step 15: Discover F from C

**Queue: D → I → G → H → J → L → F**

### Legend

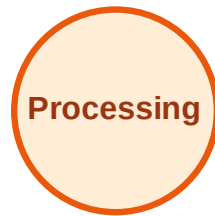
## Unseen



# BFS Traversal

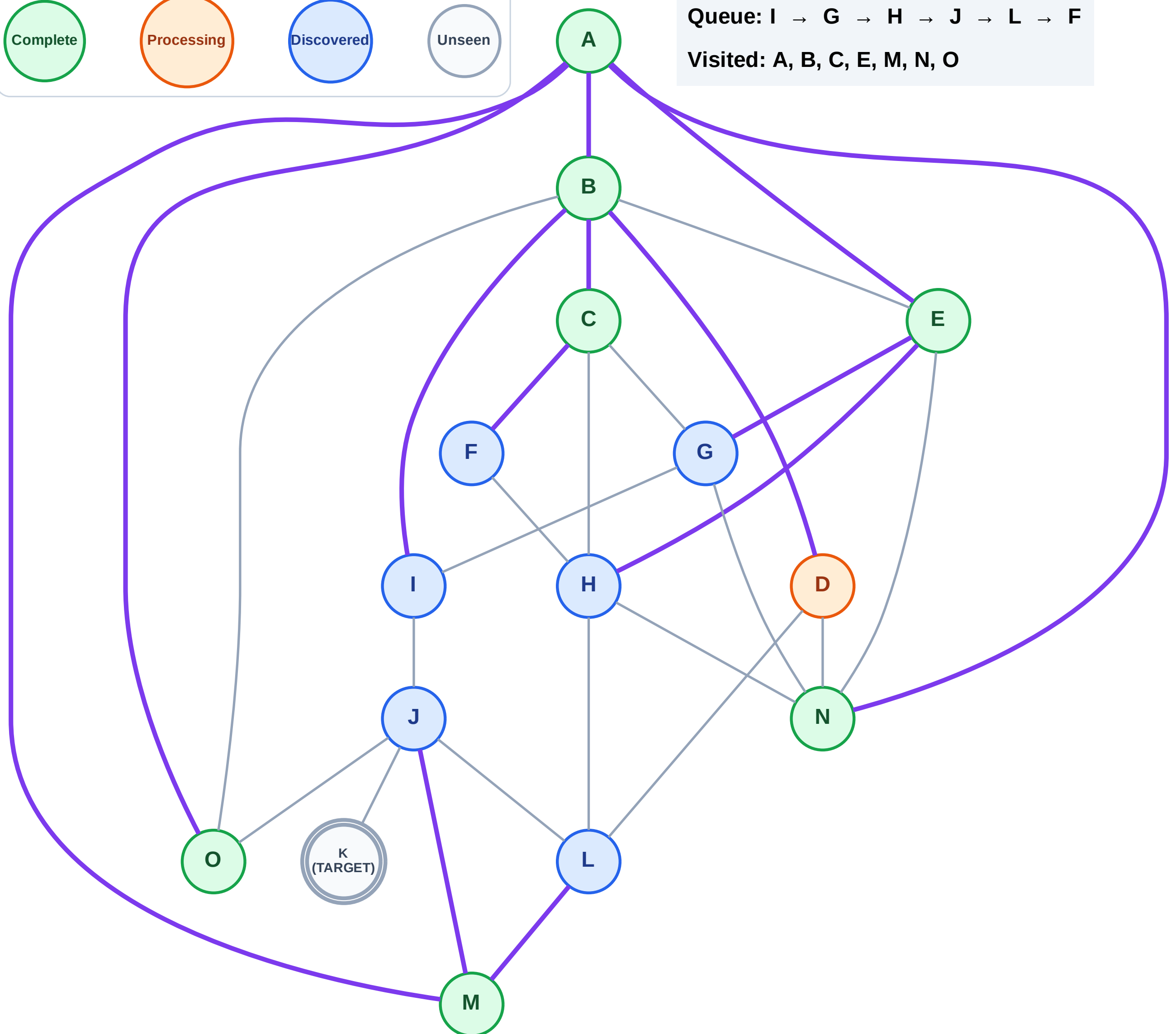
Step 16: Process node D

## Legend



Queue: I → G → H → J → L → F

Visited: A, B, C, E, M, N, O



# BFS Traversal

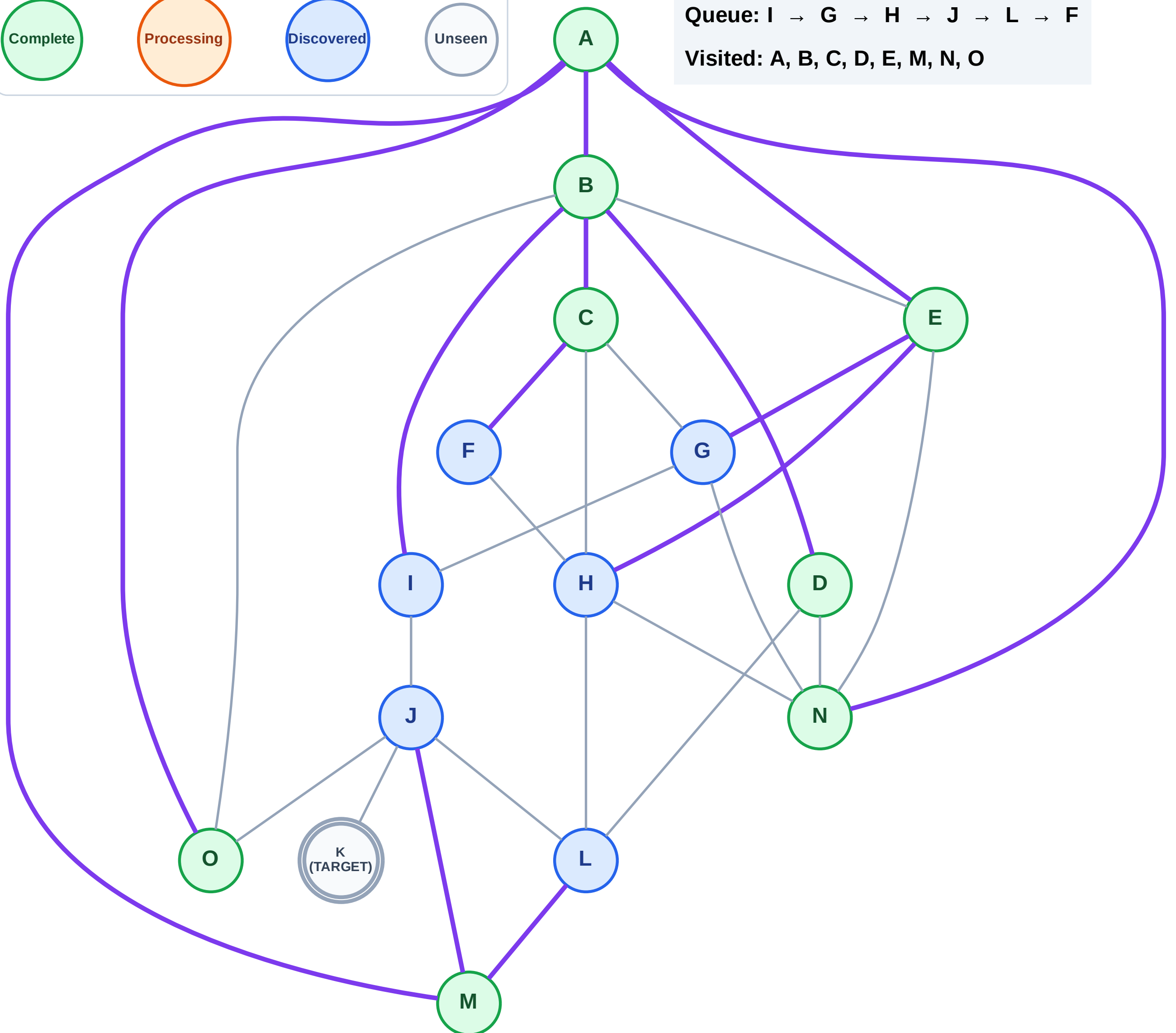
Step 17: No new neighbors from D

## Legend



Queue: I → G → H → J → L → F

Visited: A, B, C, D, E, M, N, O



# BFS Traversal

Step 18: Process node I

## Legend



Complete



Processing



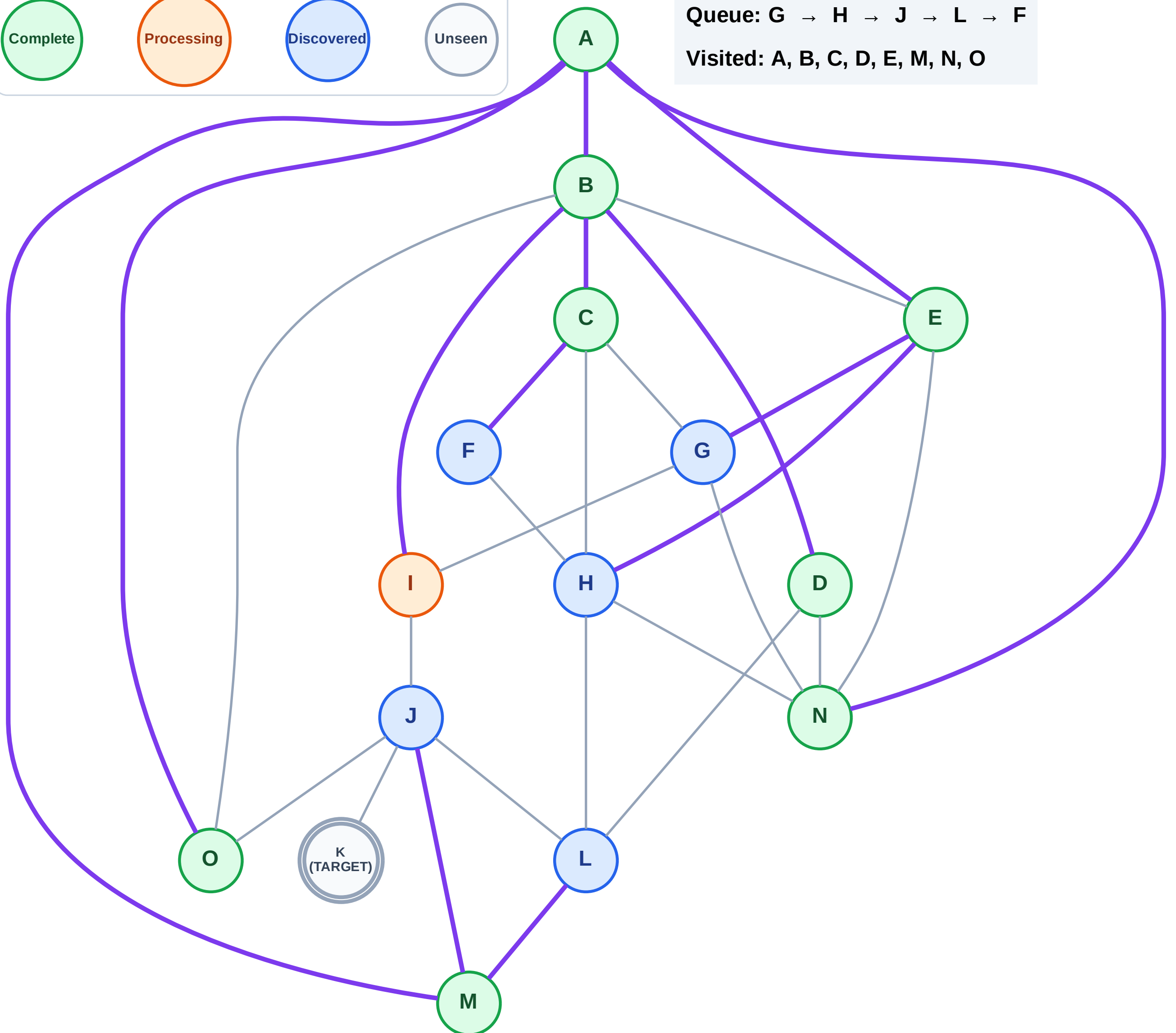
Discovered



Unseen

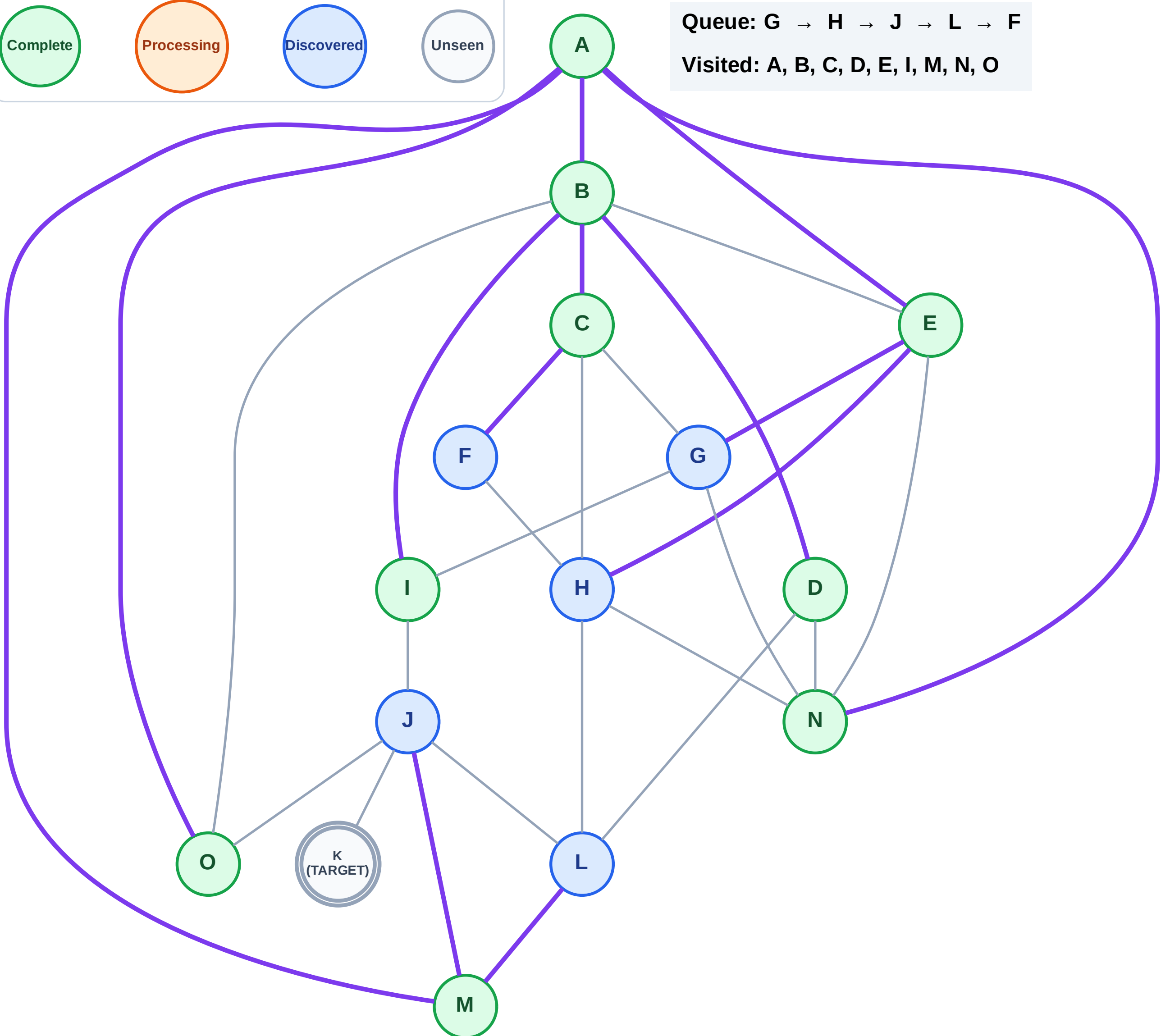
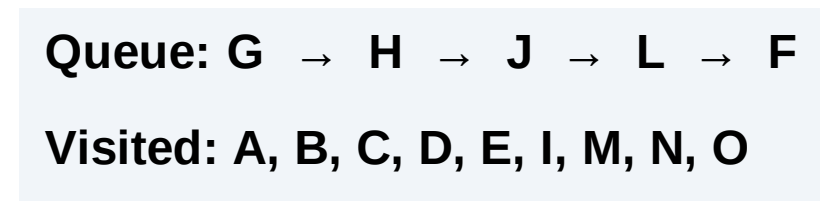
Queue: G → H → J → L → F

Visited: A, B, C, D, E, M, N, O



Step 19: No new neighbors from I

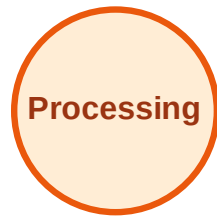
Step 19: No new neighbors from I



# BFS Traversal

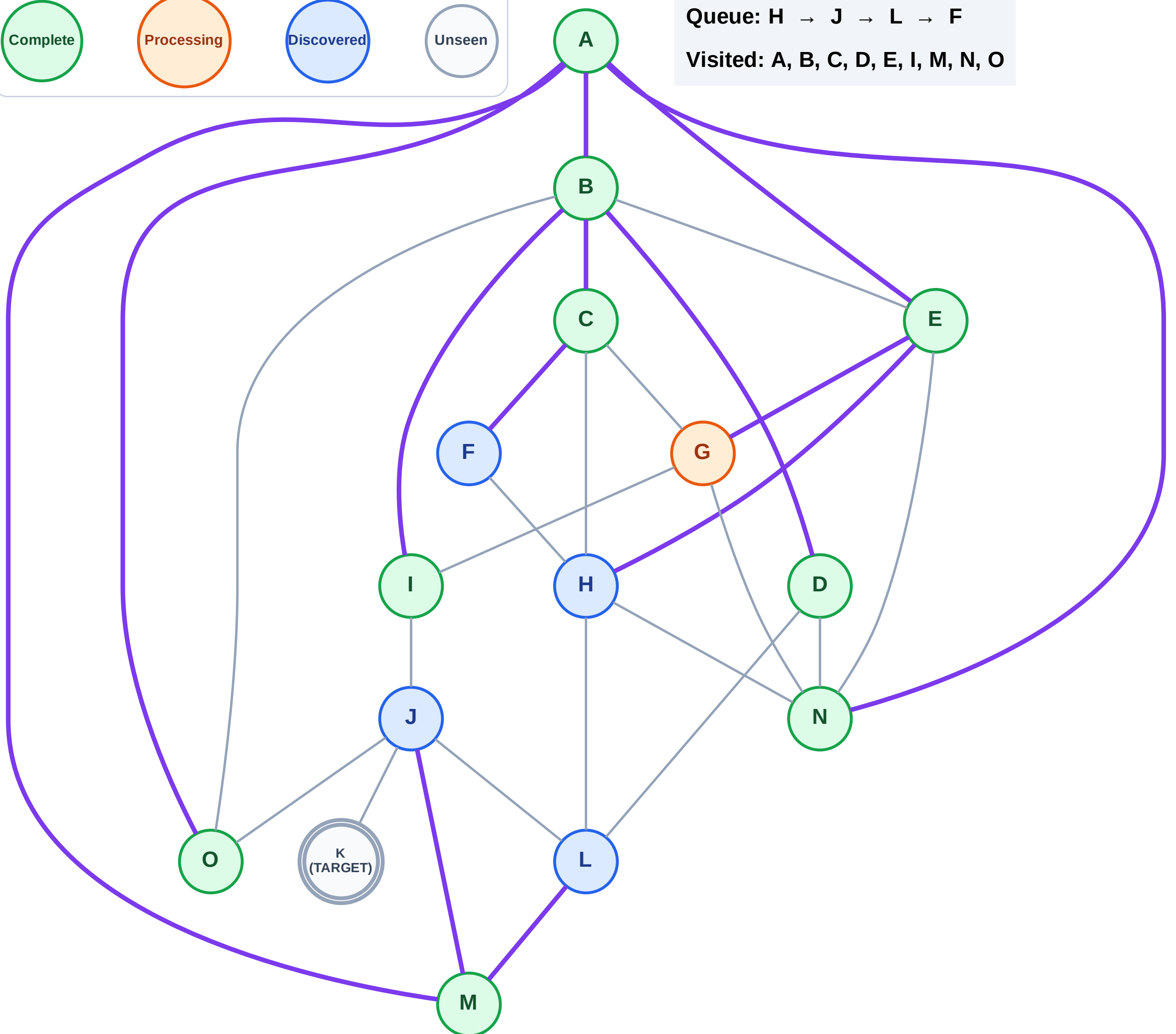
Step 20: Process node G

## Legend



Queue: H → J → L → F

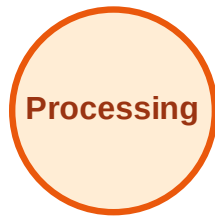
Visited: A, B, C, D, E, I, M, N, O



# BFS Traversal

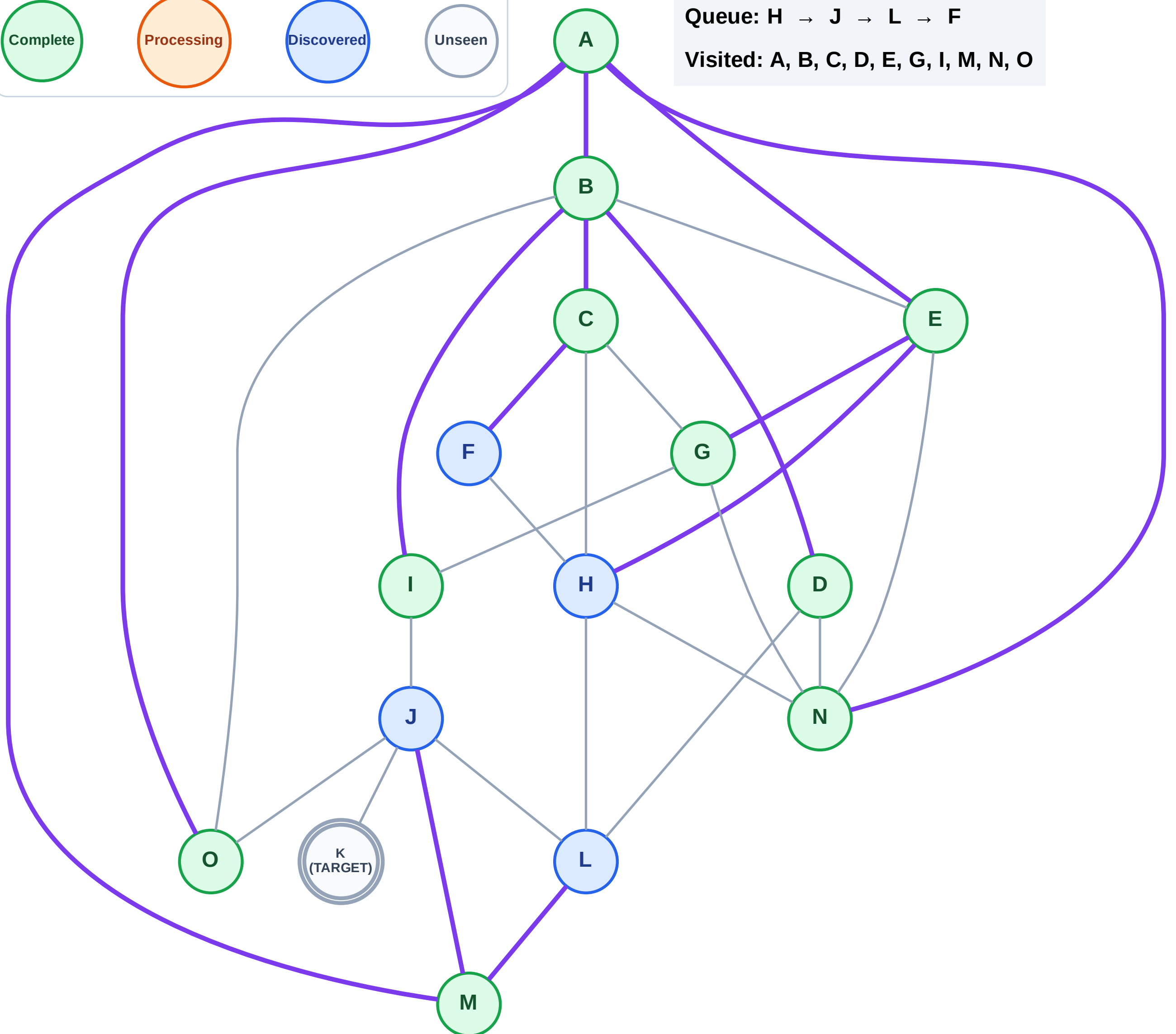
Step 21: No new neighbors from G

## Legend



Queue: H → J → L → F

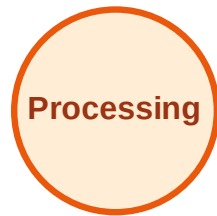
Visited: A, B, C, D, E, G, I, M, N, O



# BFS Traversal

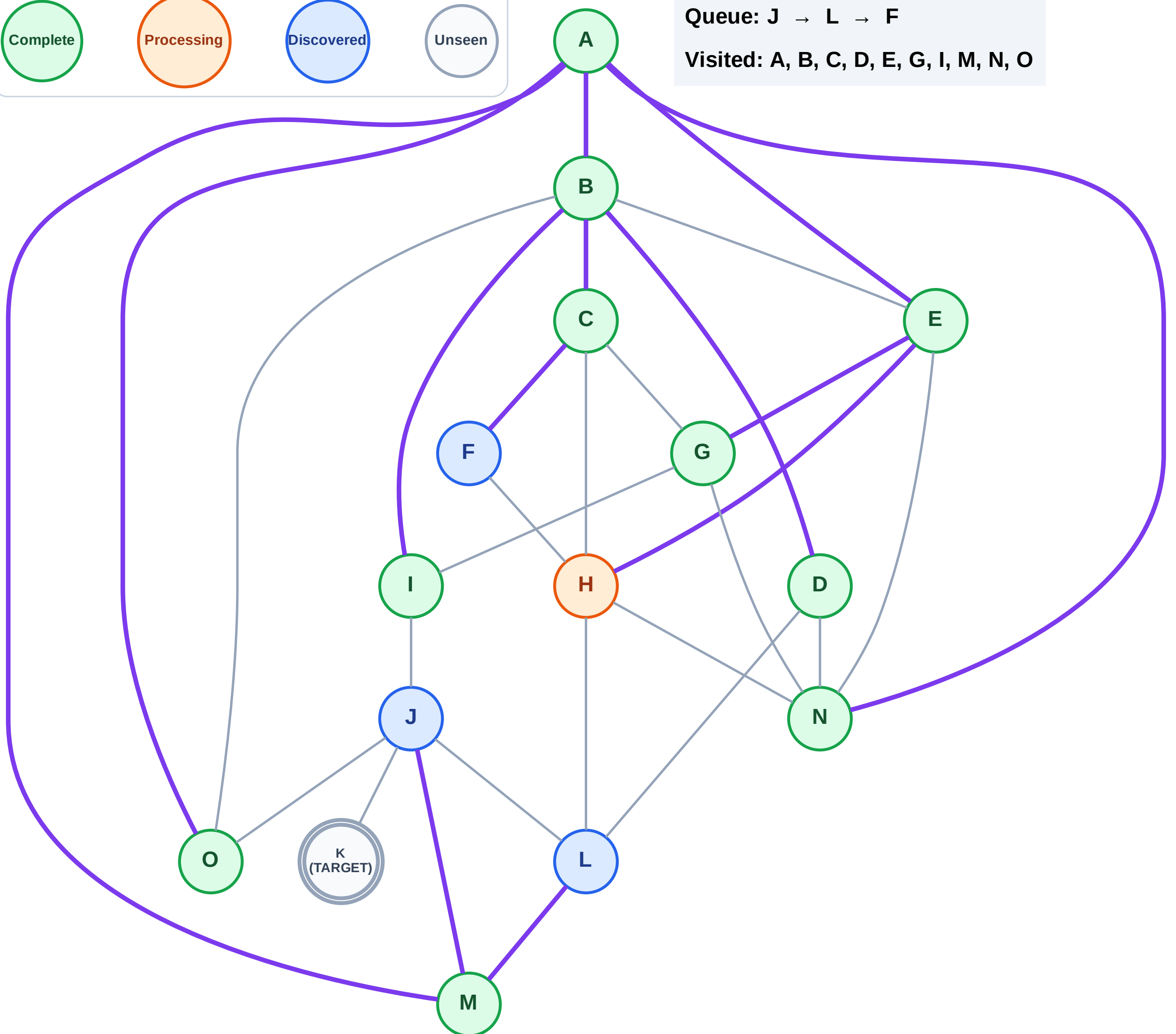
Step 22: Process node H

## Legend



Queue: J → L → F

Visited: A, B, C, D, E, G, I, M, N, O



Step 23: No new neighbors from H

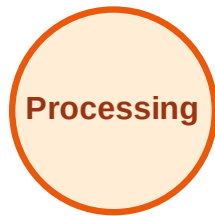
Step 23: No new neighbors from H



# BFS Traversal

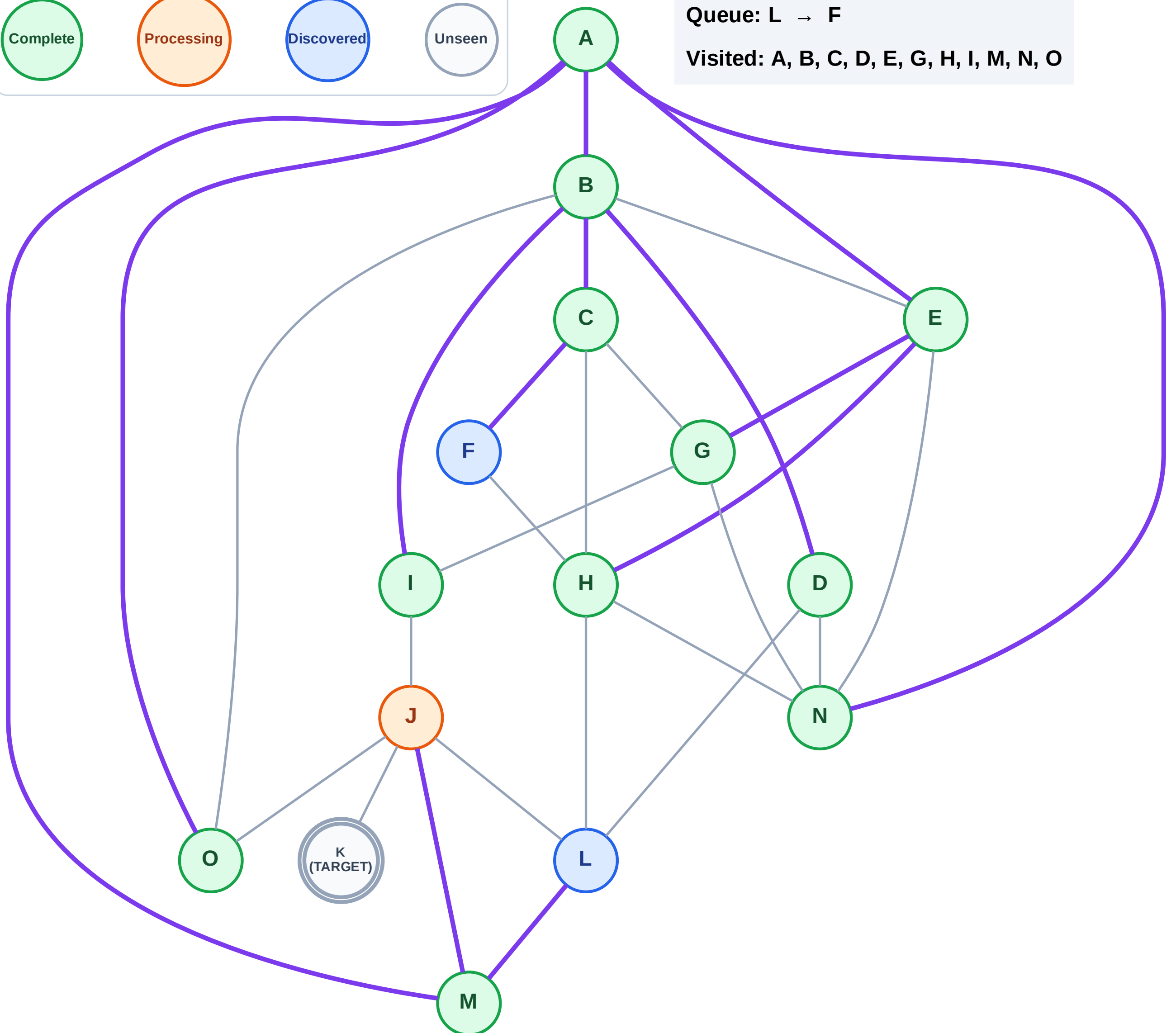
Step 24: Process node J

## Legend

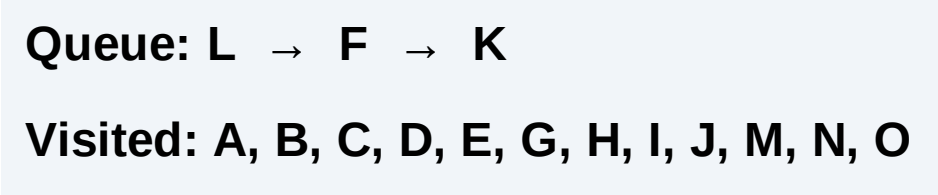


Queue: L → F

Visited: A, B, C, D, E, G, H, I, M, N, O



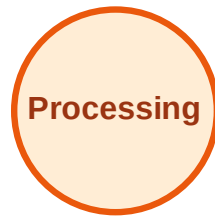
## Step 25: Discover K from J



# BFS Traversal

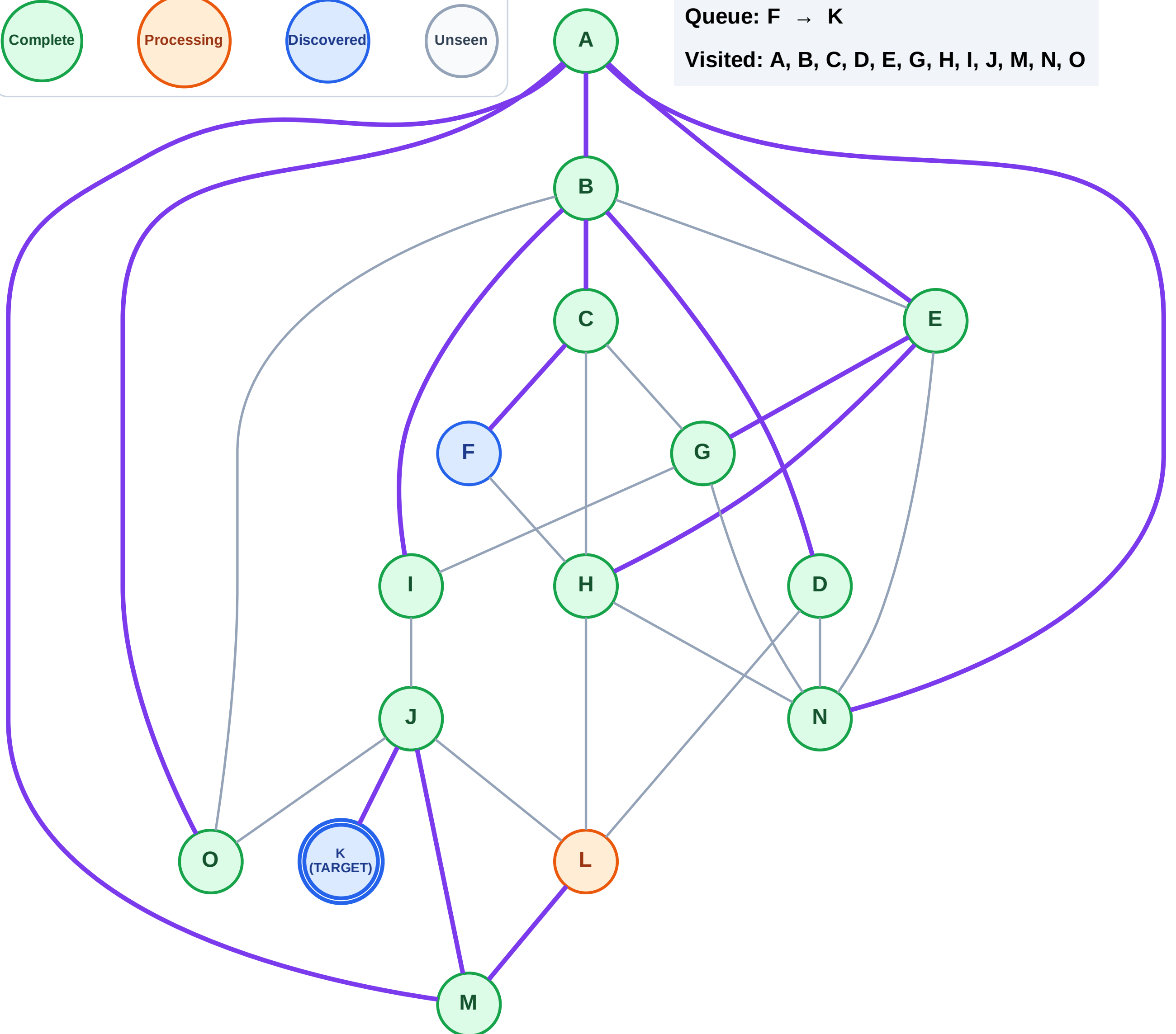
Step 26: Process node L

## Legend



Queue: F → K

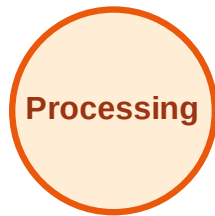
Visited: A, B, C, D, E, G, H, I, J, M, N, O



# BFS Traversal

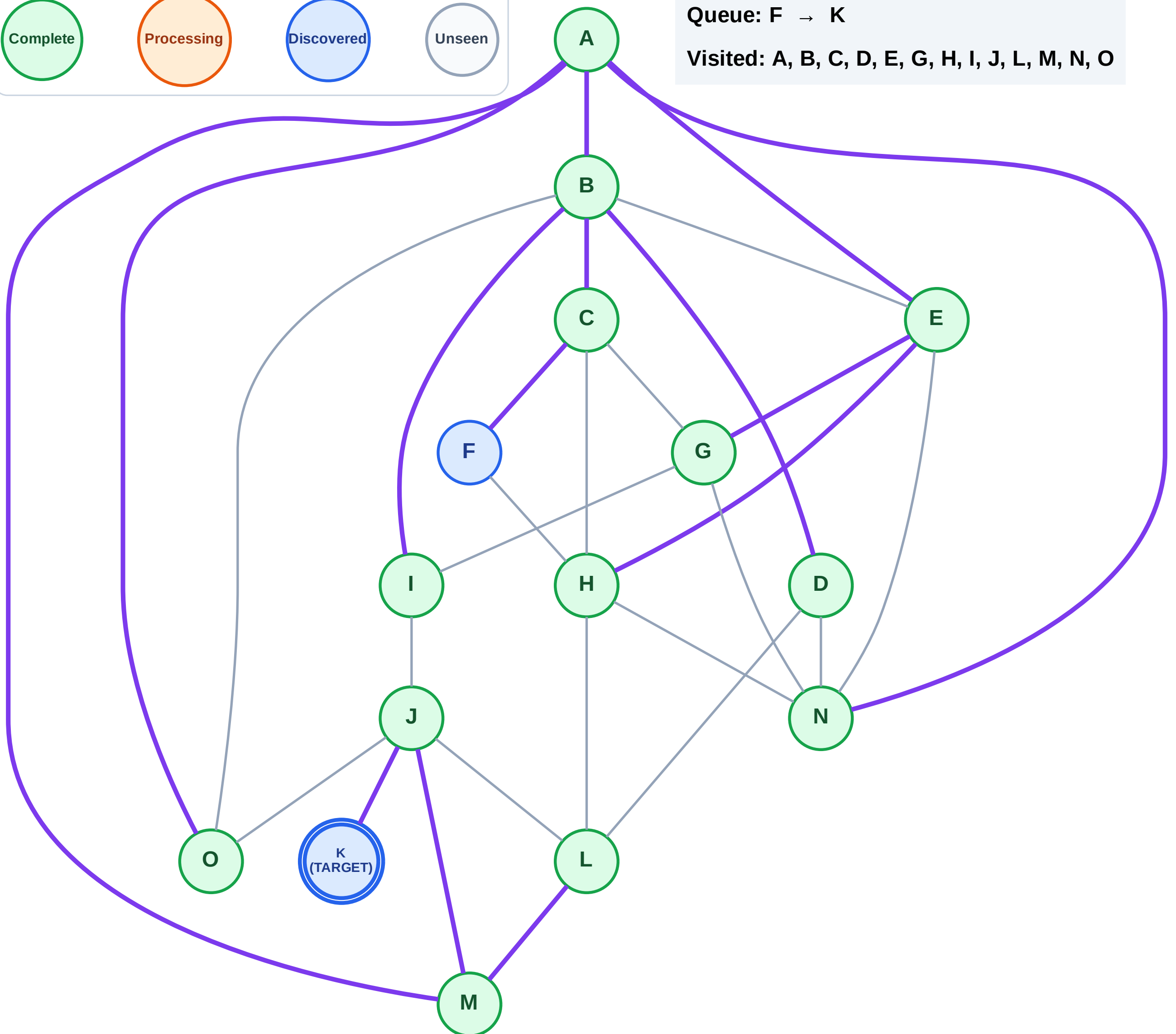
Step 27: No new neighbors from L

## Legend



Queue: F → K

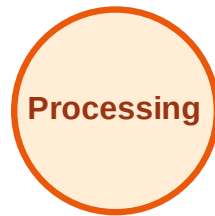
Visited: A, B, C, D, E, G, H, I, J, L, M, N, O



# BFS Traversal

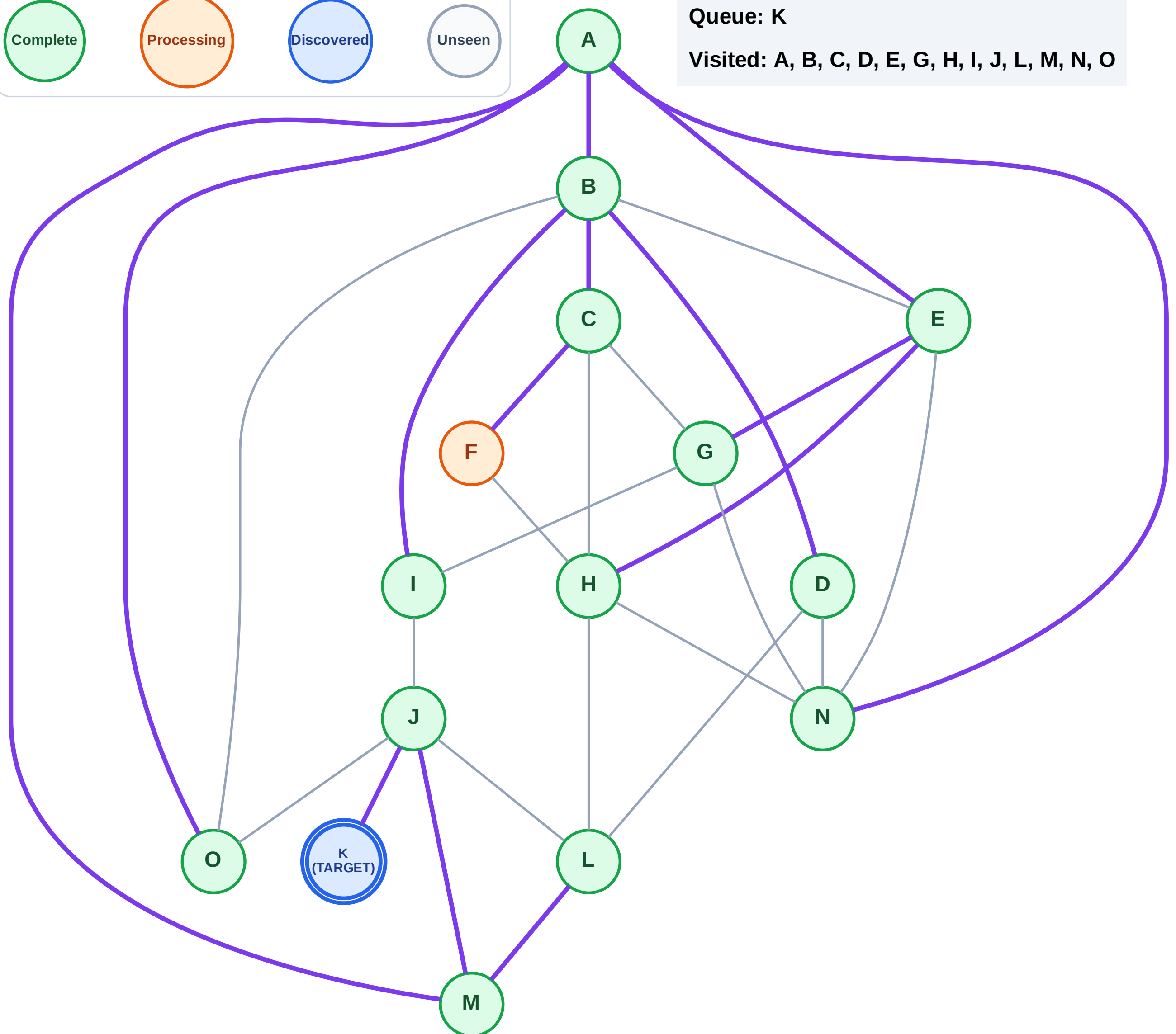
Step 28: Process node F

## Legend



Queue: K

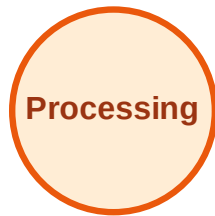
Visited: A, B, C, D, E, G, H, I, J, L, M, N, O



# BFS Traversal

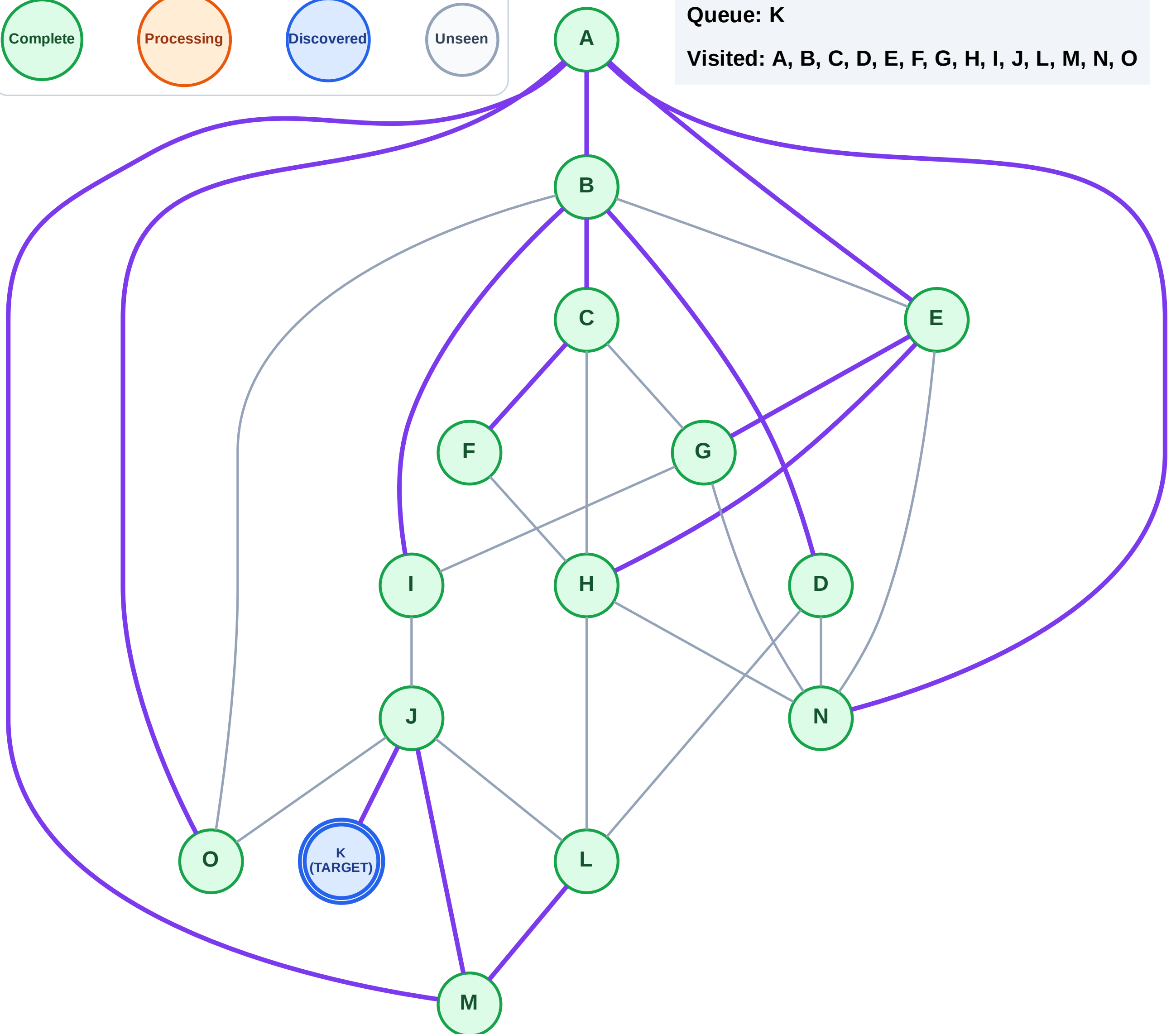
Step 29: No new neighbors from F

## Legend



Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



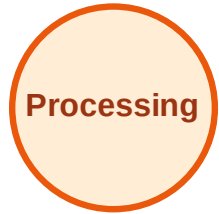
# BFS Traversal

Step 30: Process node K

## Legend



Complete



Processing



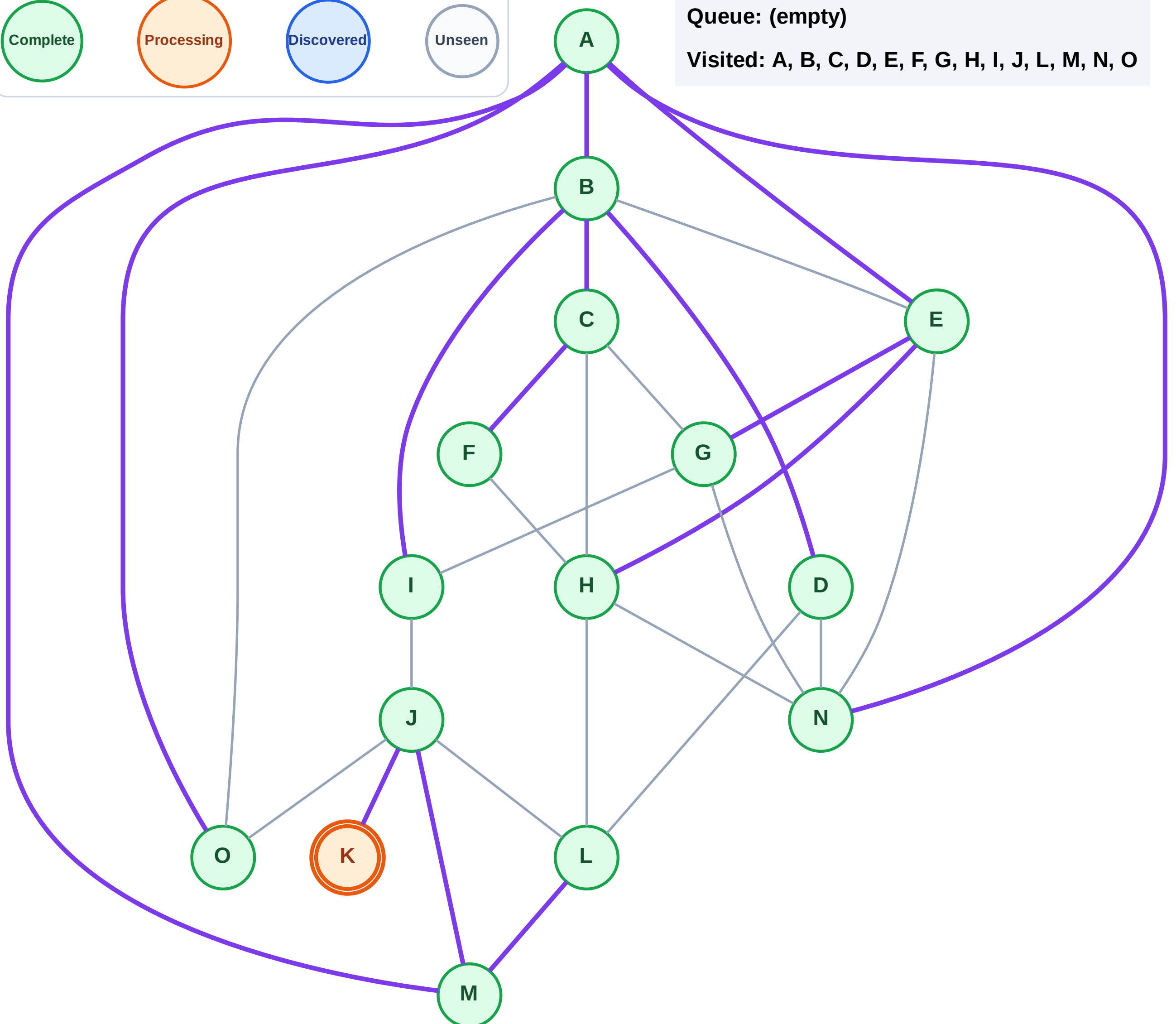
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



# BFS Traversal

Step 31: TARGET NODE K REACHED! Path (3 hops): A → M → J → K

Legend

Complete

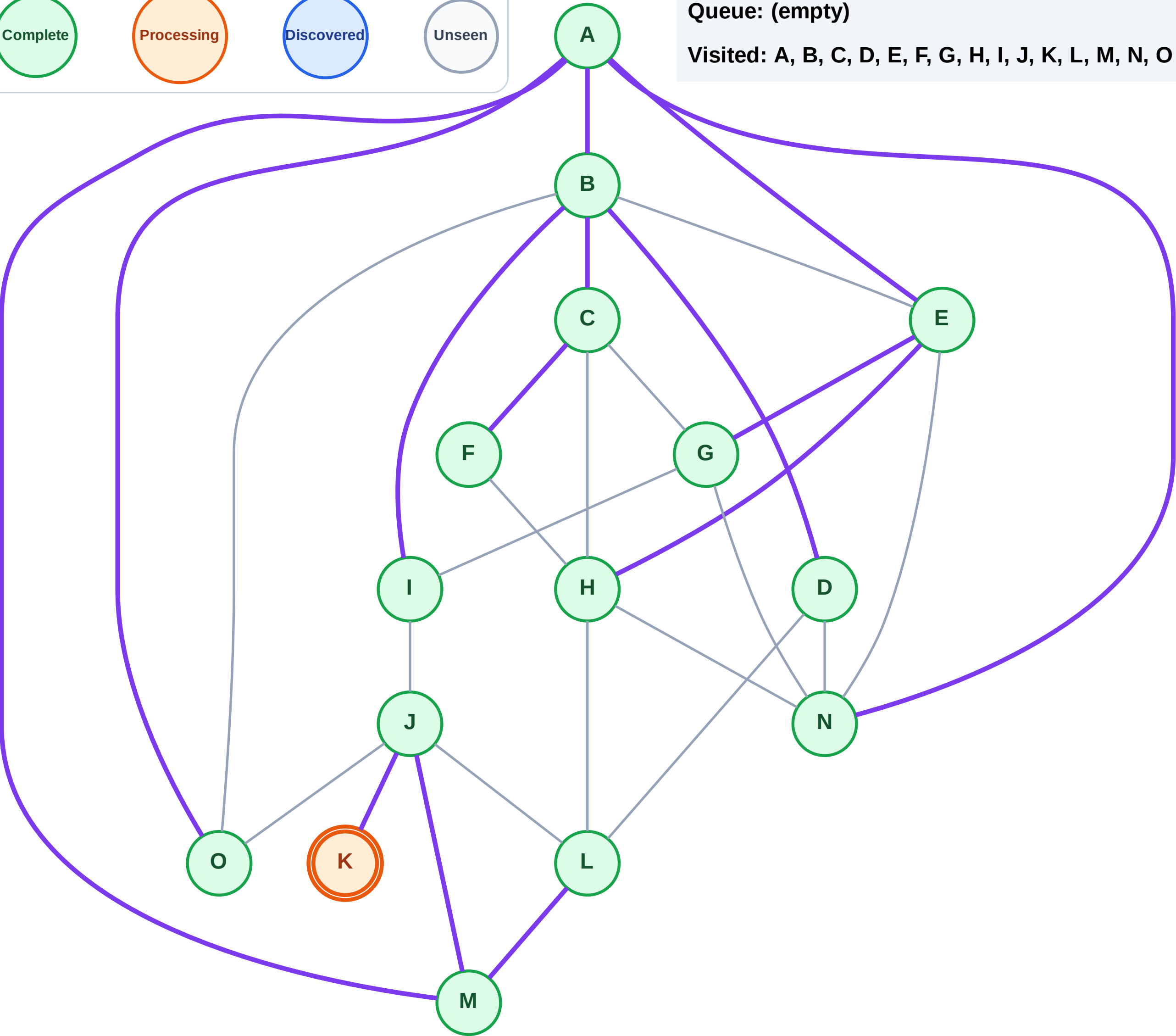
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



# DFS Traversal

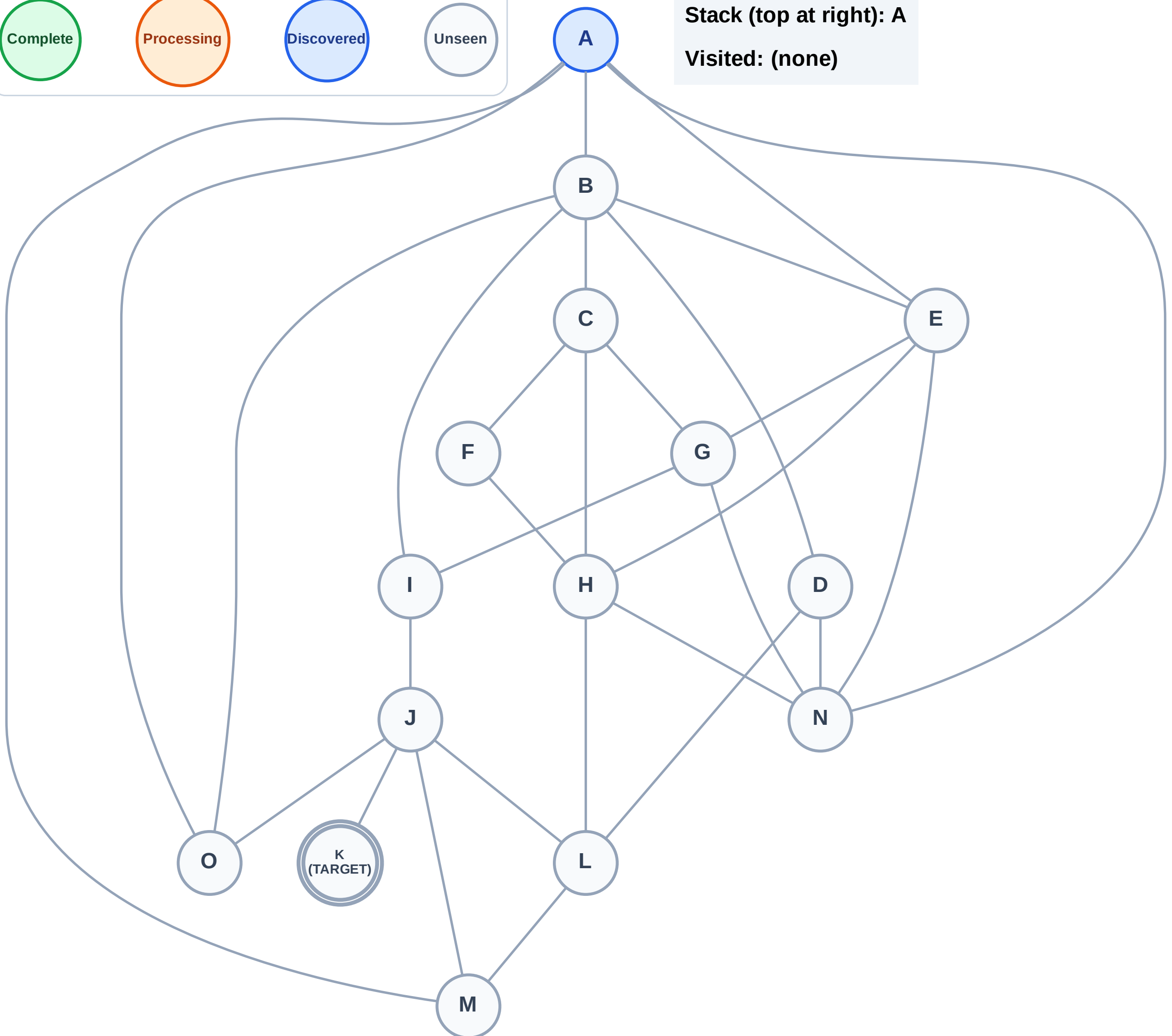
Step 1: Start at A (Seeking target node K)

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

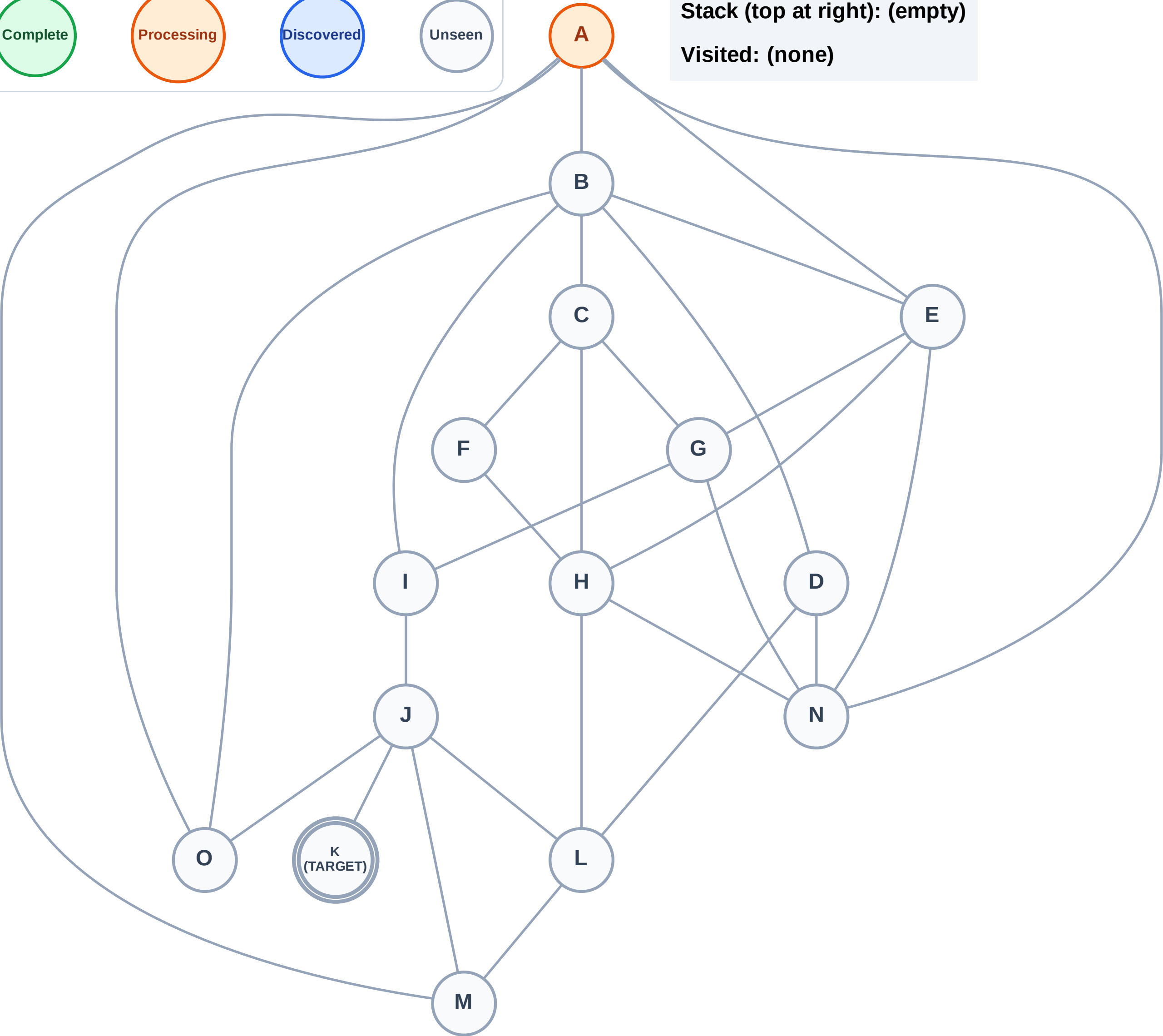
Complete

Processing

Discovered

Unseen

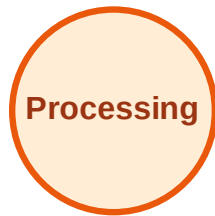
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

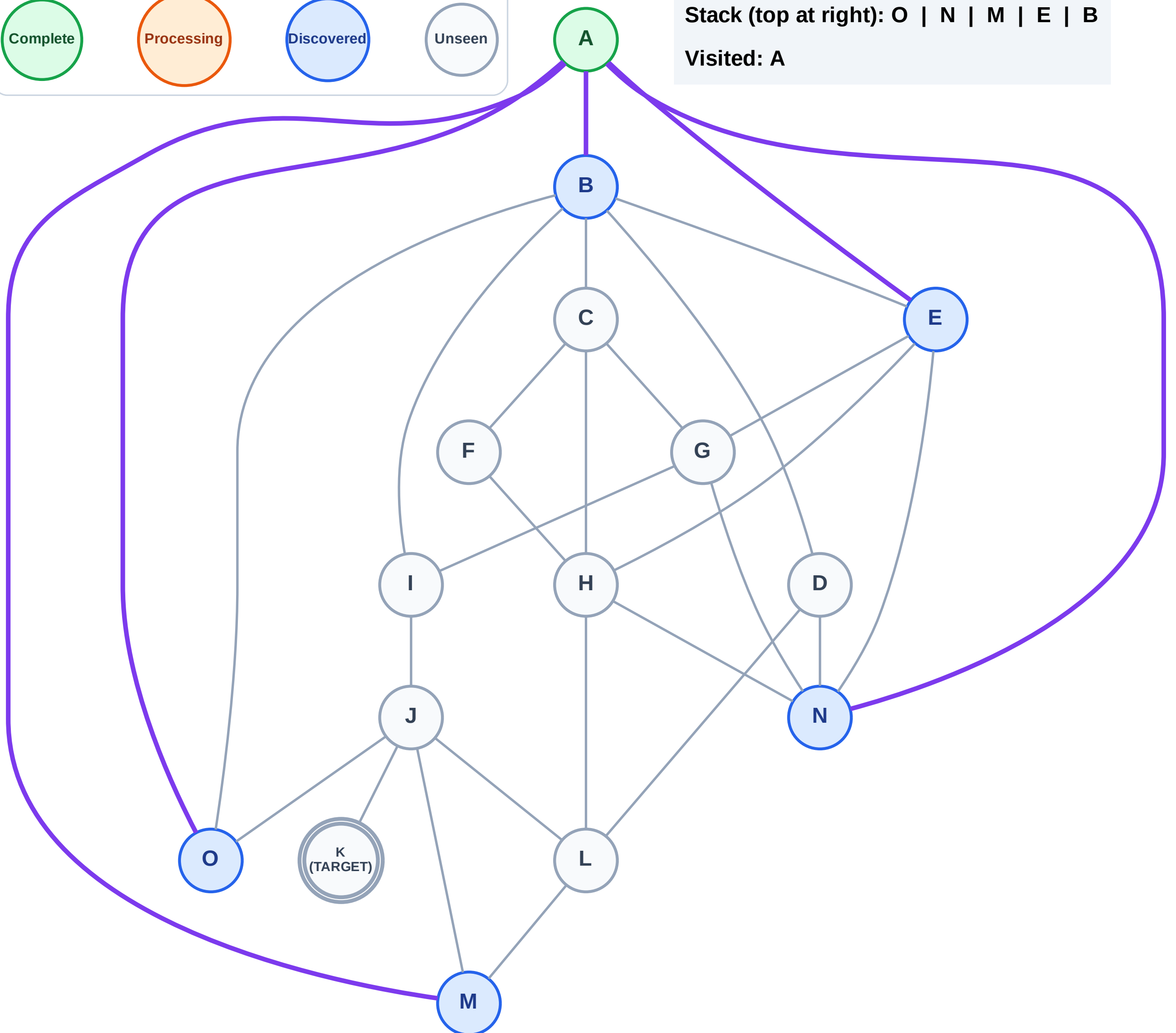
Step 3: Discover O, N, M, E, B from A

## Legend



Stack (top at right): O | N | M | E | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

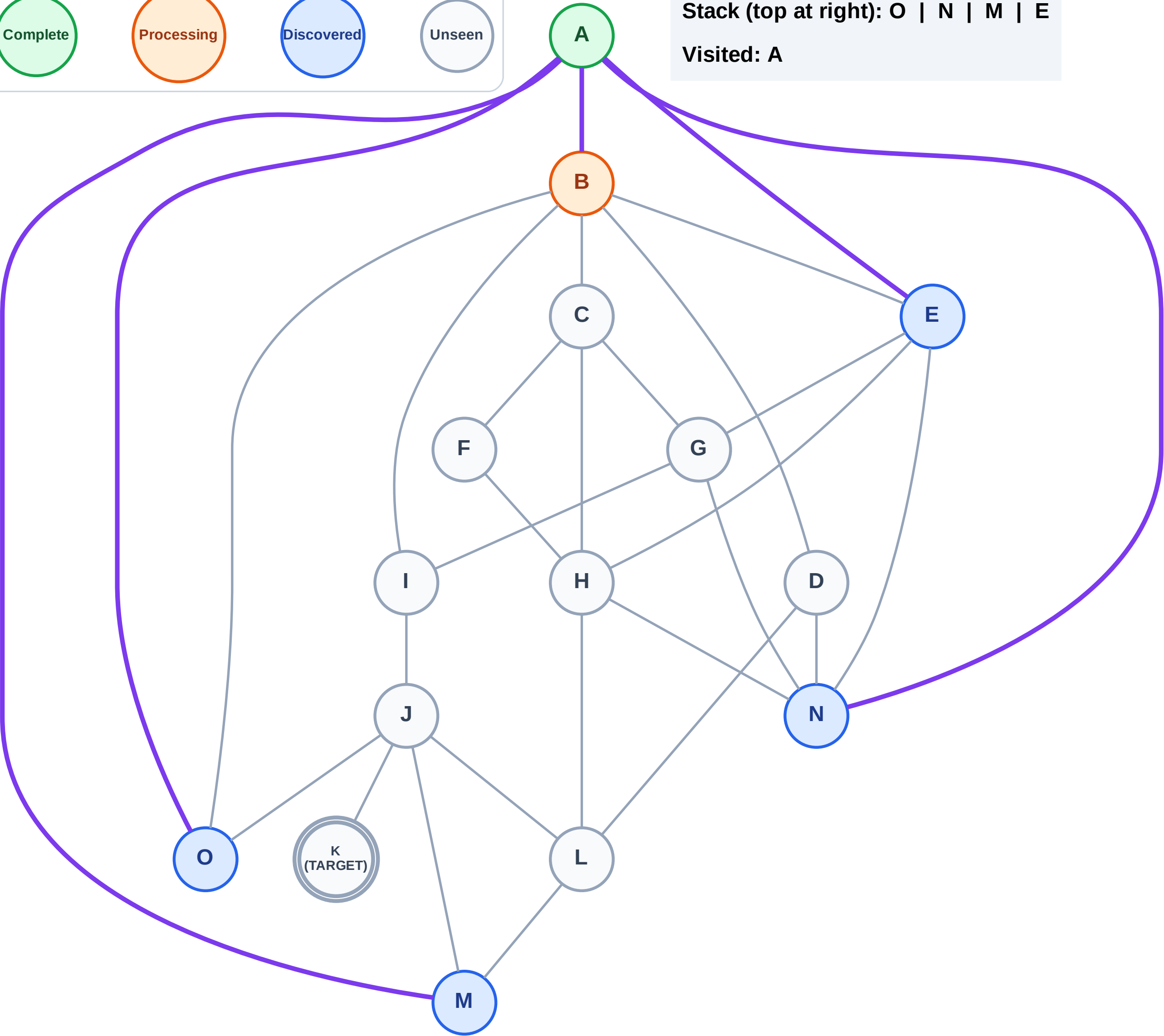
Processing

Discovered

Unseen

Stack (top at right): O | N | M | E

Visited: A



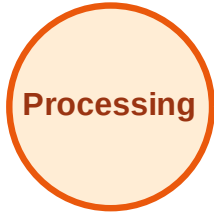
# DFS Traversal

Step 5: Discover I, D, C from B

## Legend



Complete



Processing



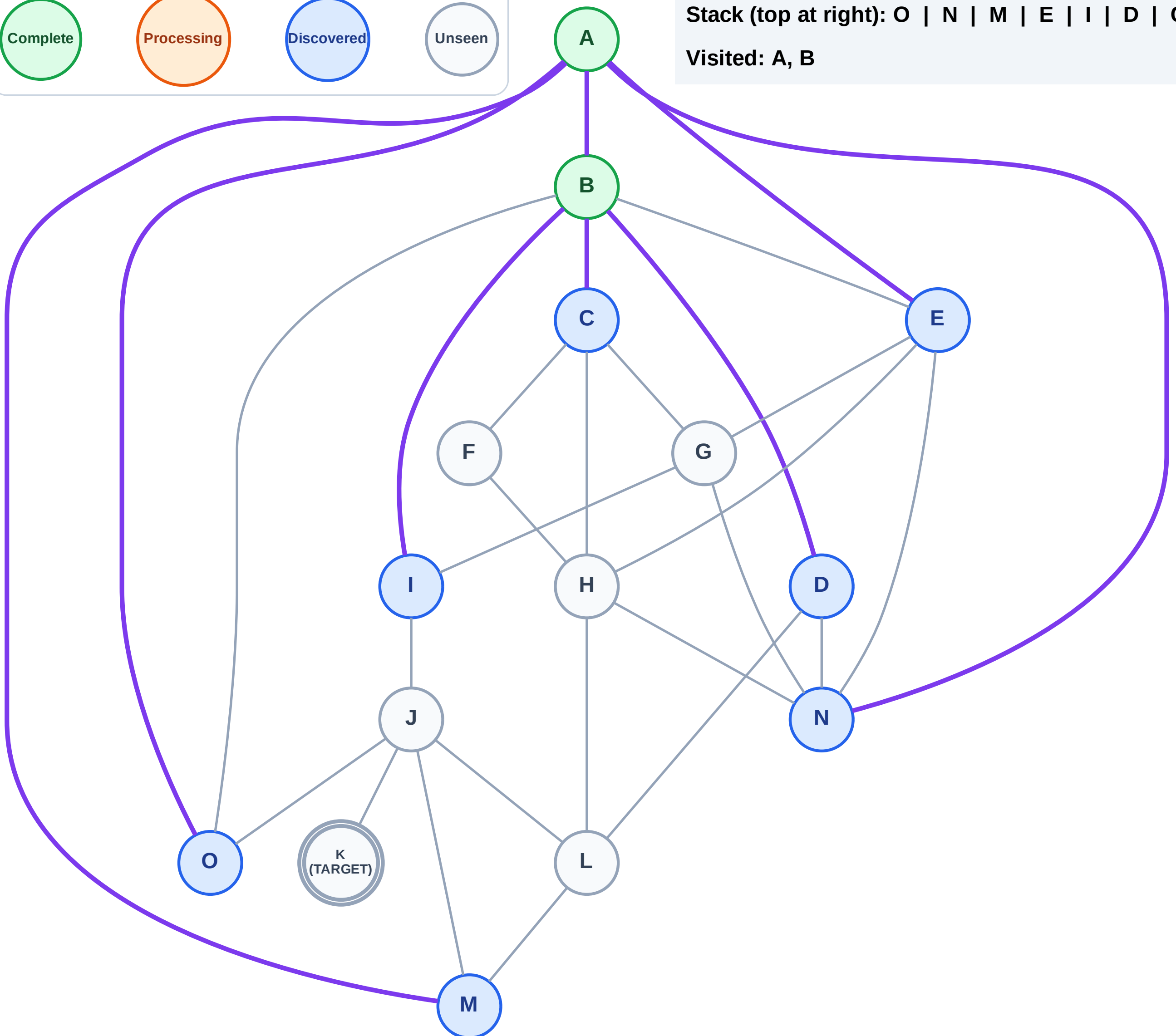
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D | C

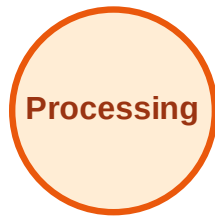
Visited: A, B



# DFS Traversal

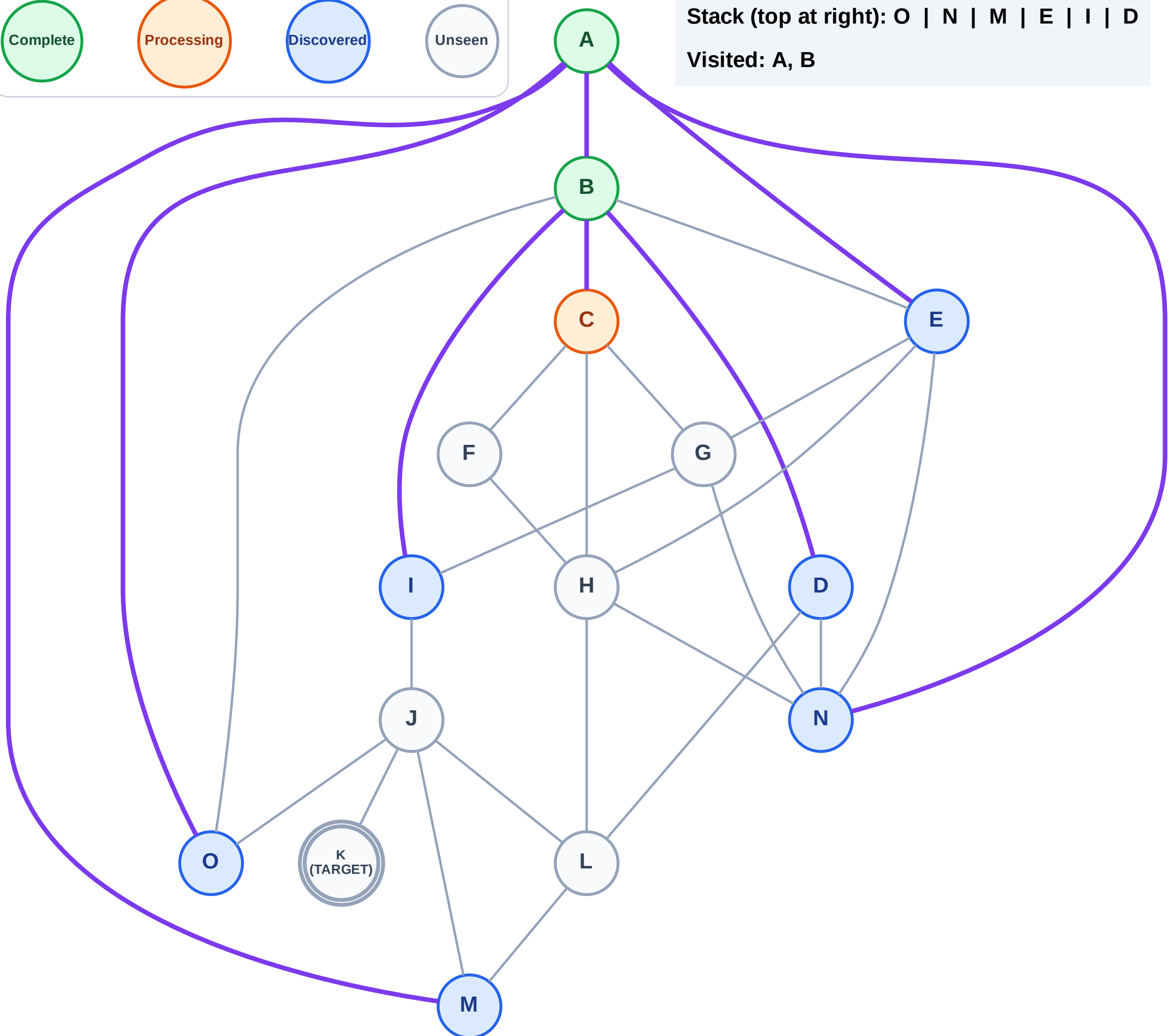
Step 6: Process node C

## Legend



Stack (top at right): O | N | M | E | I | D

Visited: A, B

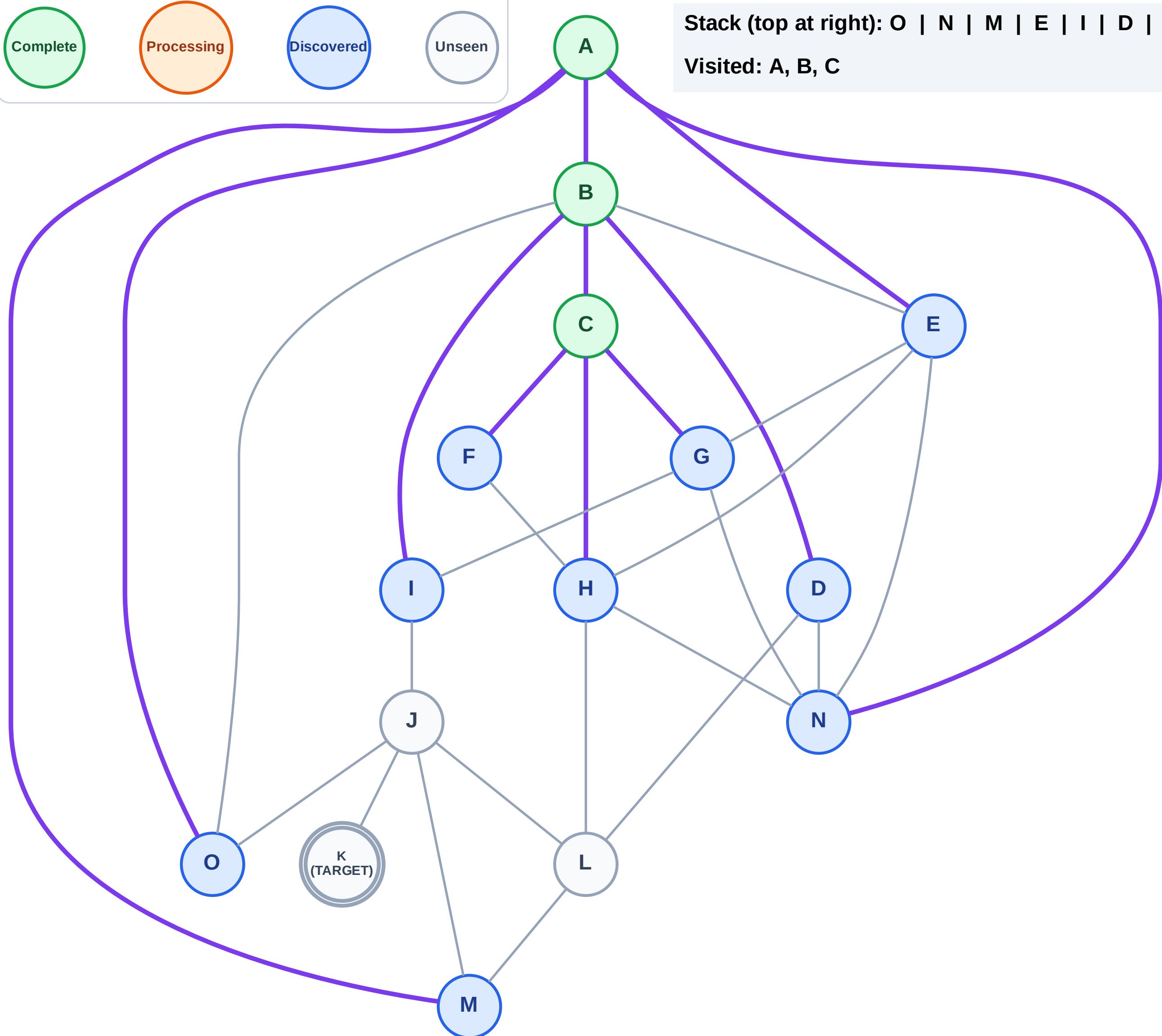


### Step 7: Discover H, G, F from C

**Stack (top at right):** O | N | M | E | I | D | H | G | F

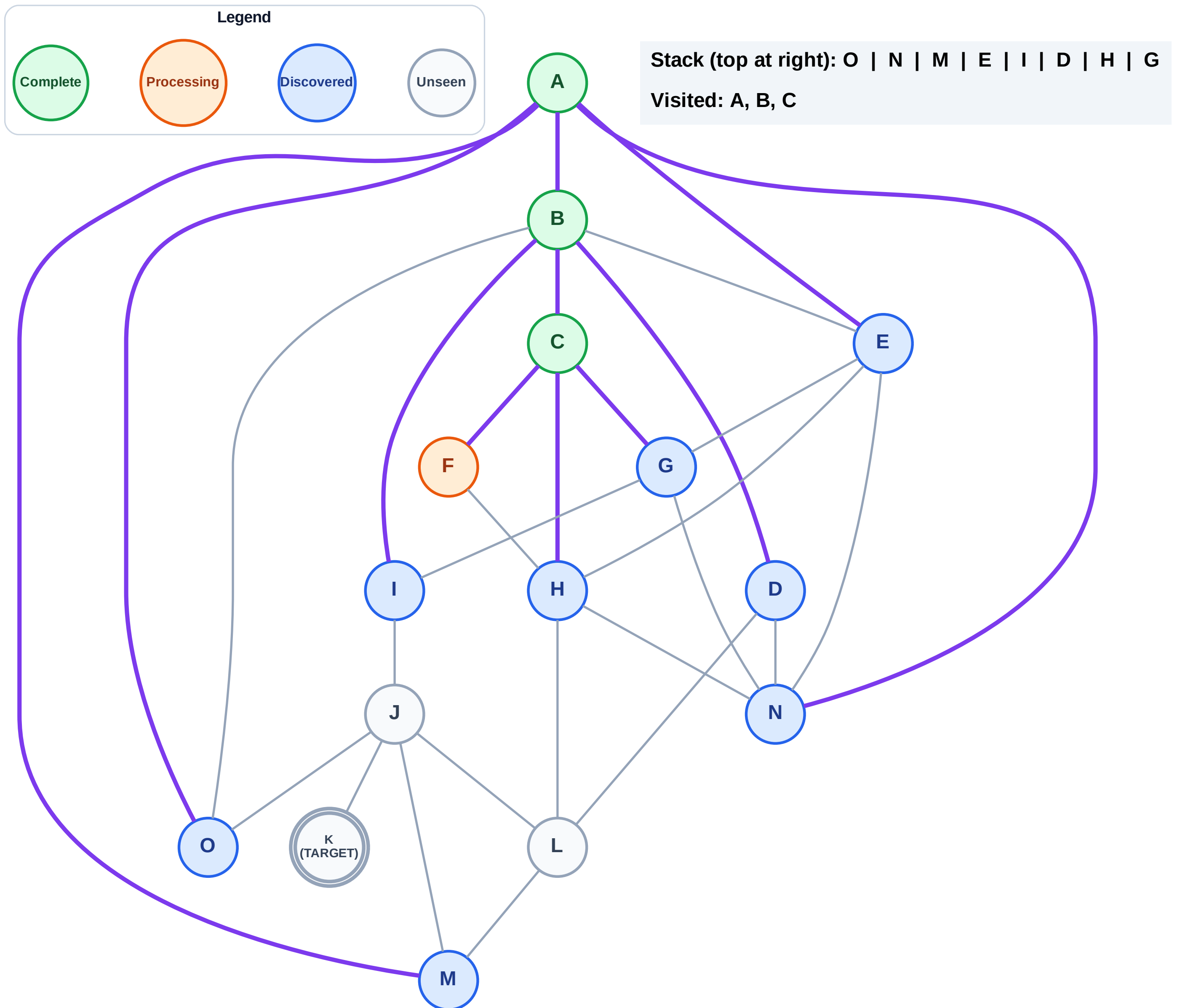
### Legend

## Unseen



### Step 8: Process node F

### Step 8: Process node F

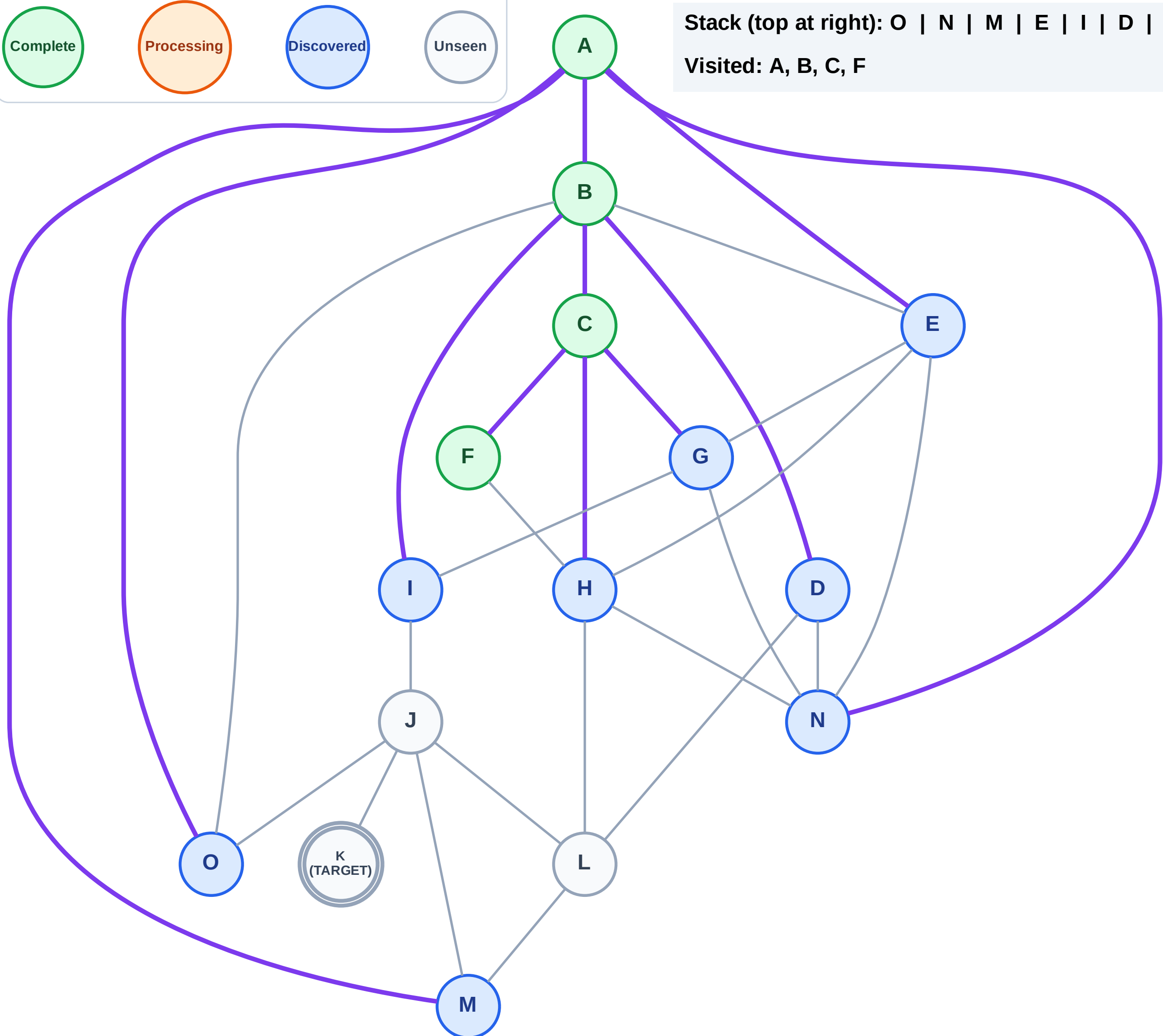


Step 9: No new neighbors from F

**Stack (top at right):** O | N | M | E | I | D | H | G

### Legend

## Unseen



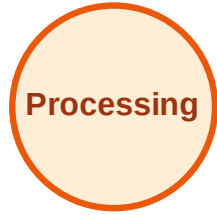
# DFS Traversal

Step 10: Process node G

## Legend



Complete



Processing



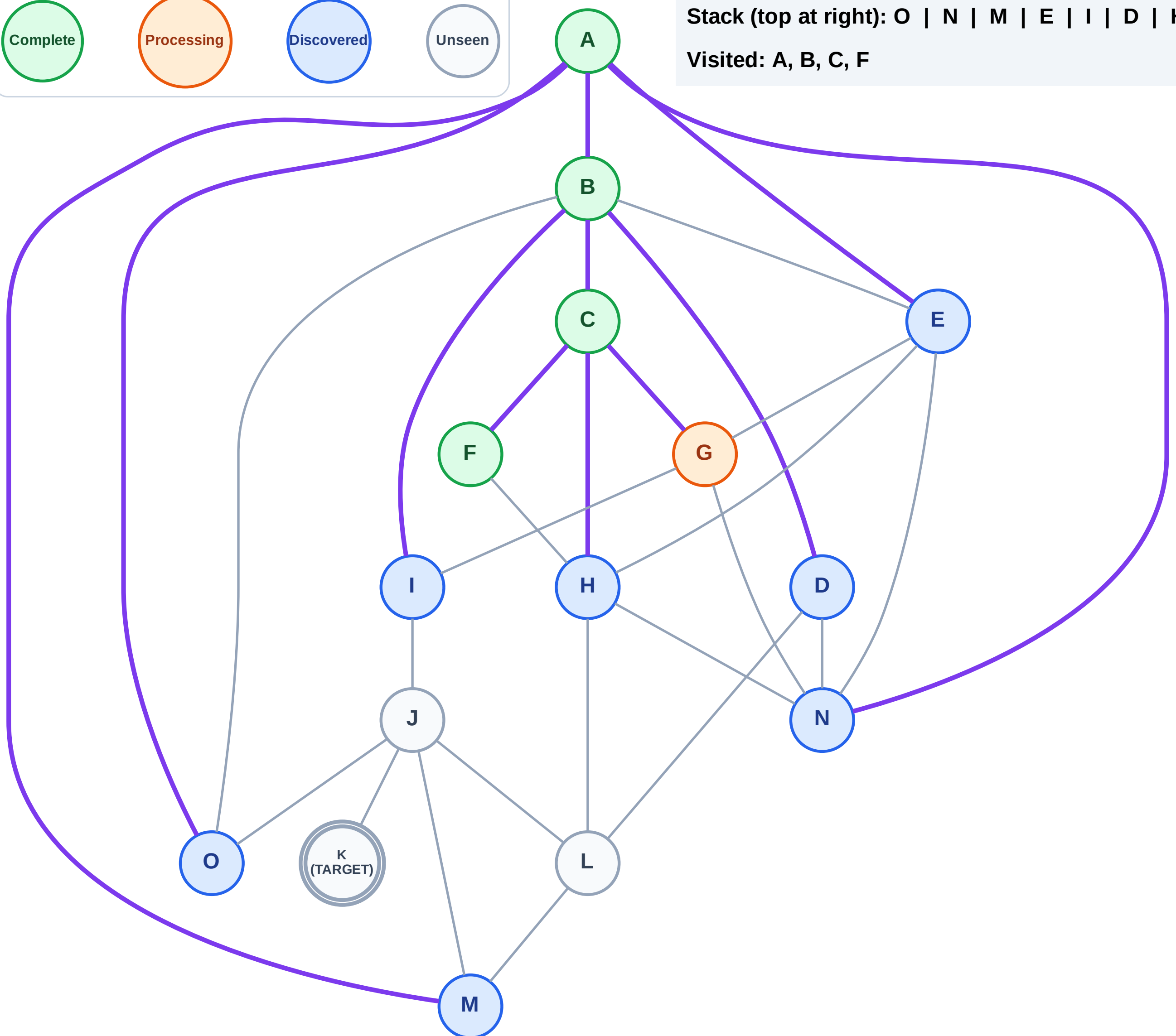
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D | H

Visited: A, B, C, F



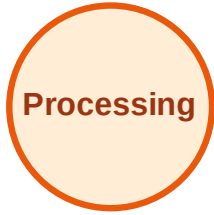
# DFS Traversal

Step 11: No new neighbors from G

## Legend



Complete



Processing



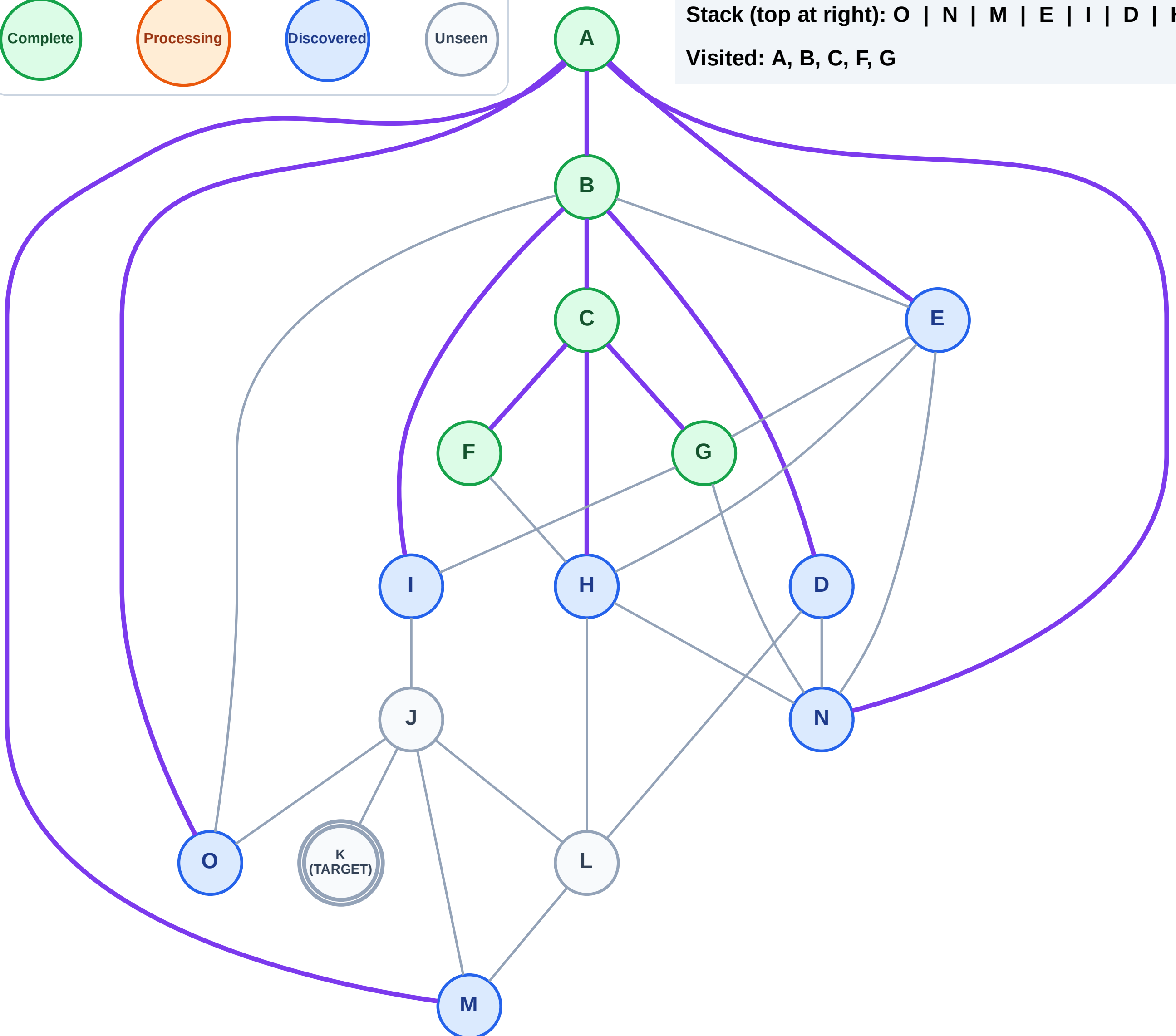
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D | H

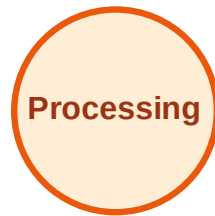
Visited: A, B, C, F, G



# DFS Traversal

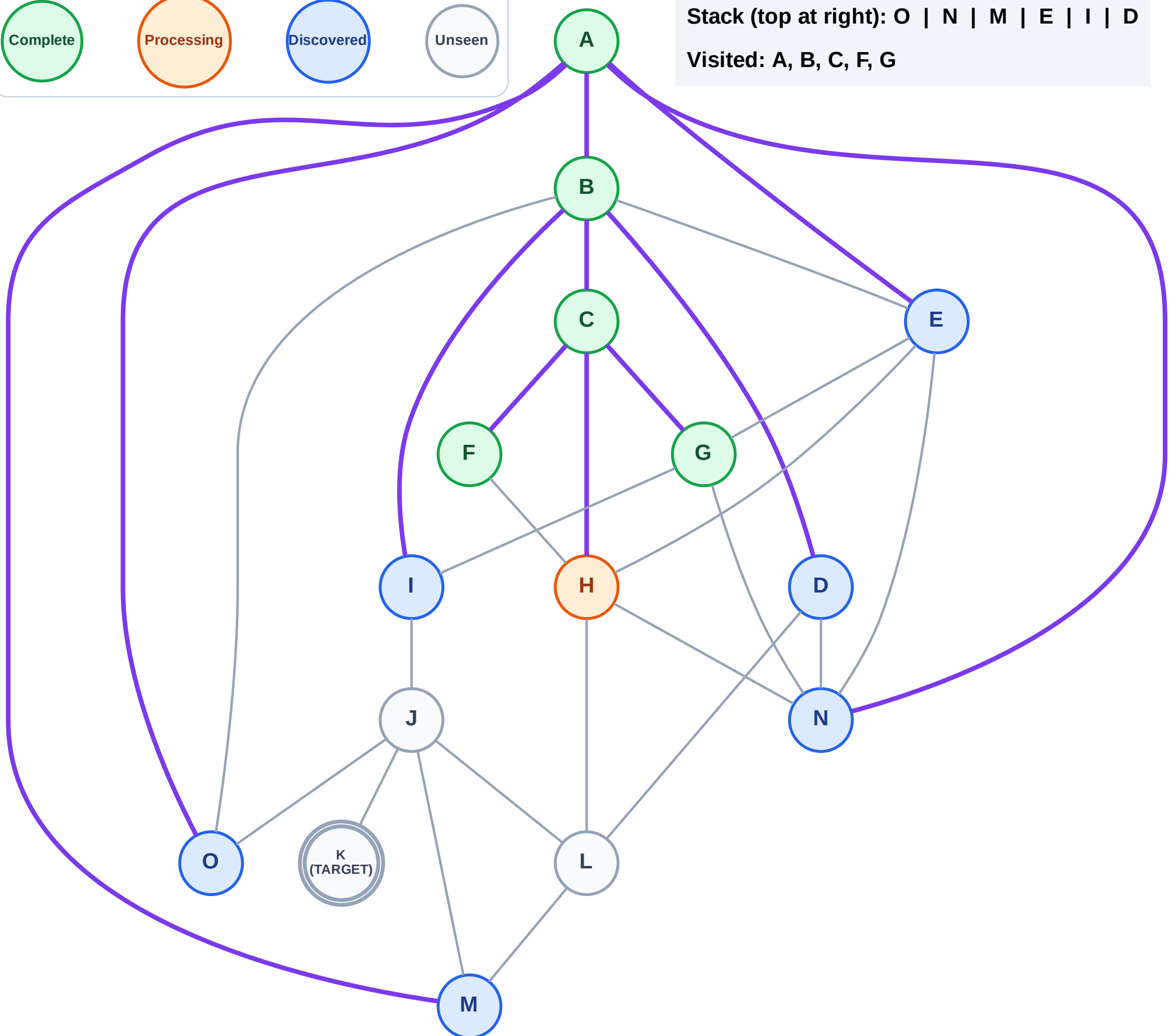
Step 12: Process node H

## Legend



Stack (top at right): O | N | M | E | I | D

Visited: A, B, C, F, G



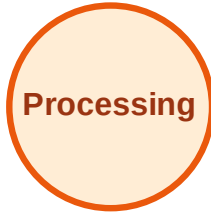
# DFS Traversal

Step 13: Discover L from H

## Legend



Complete



Processing



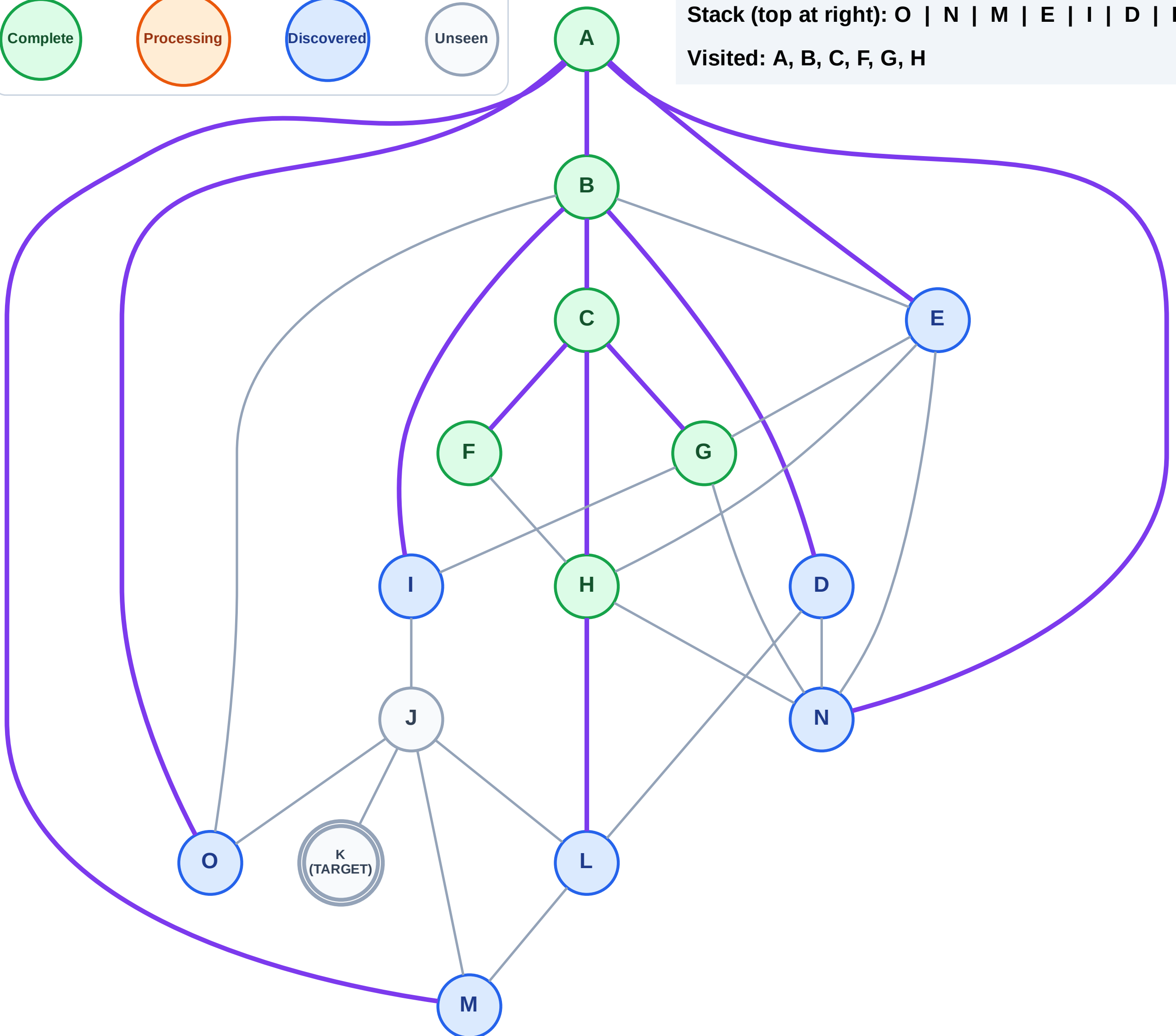
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D | L

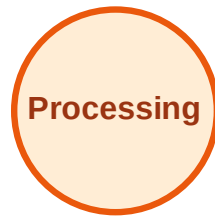
Visited: A, B, C, F, G, H



# DFS Traversal

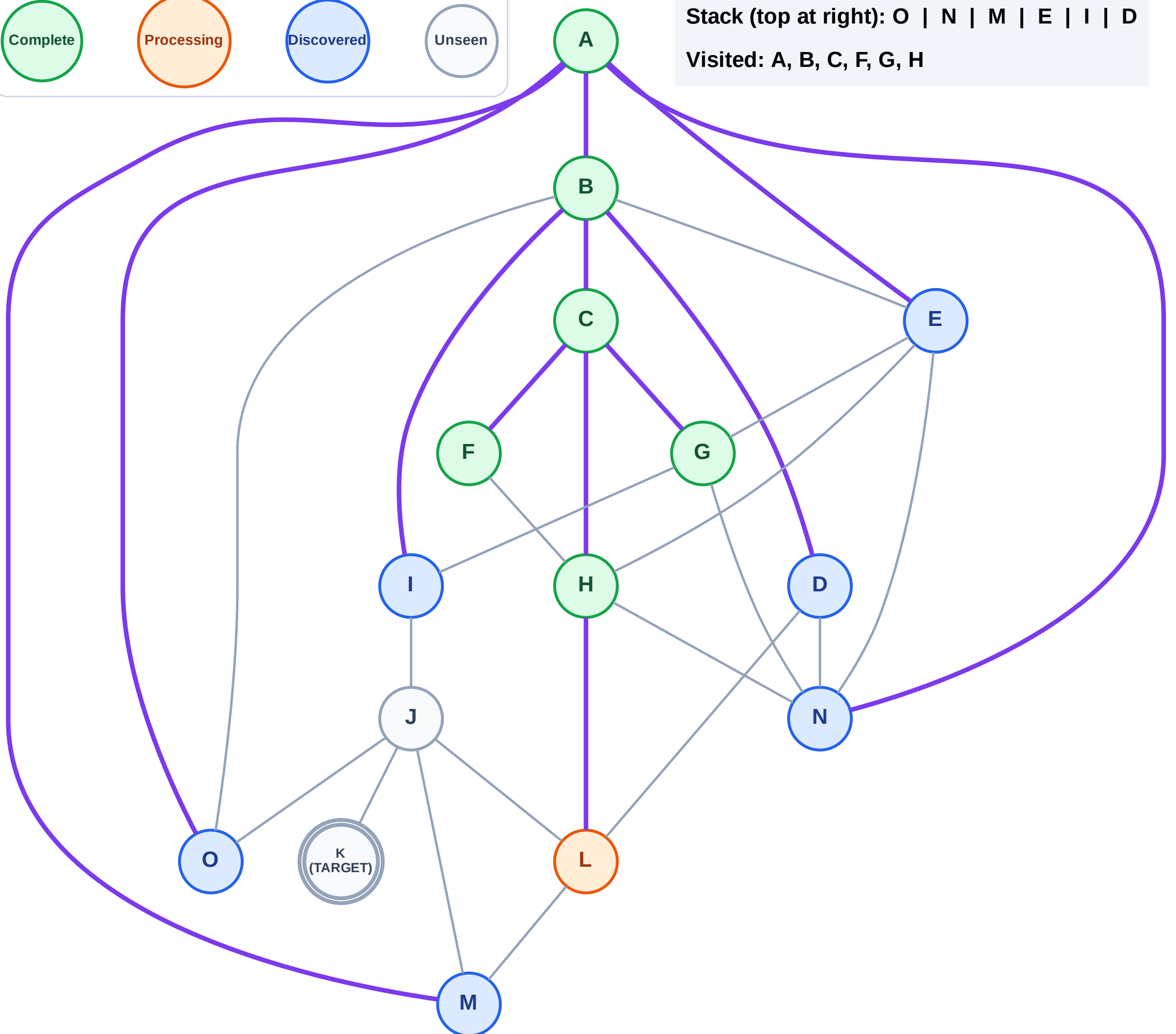
Step 14: Process node L

## Legend



Stack (top at right): O | N | M | E | I | D

Visited: A, B, C, F, G, H



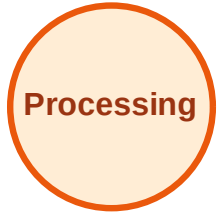
# DFS Traversal

Step 15: Discover J from L

## Legend



Complete



Processing



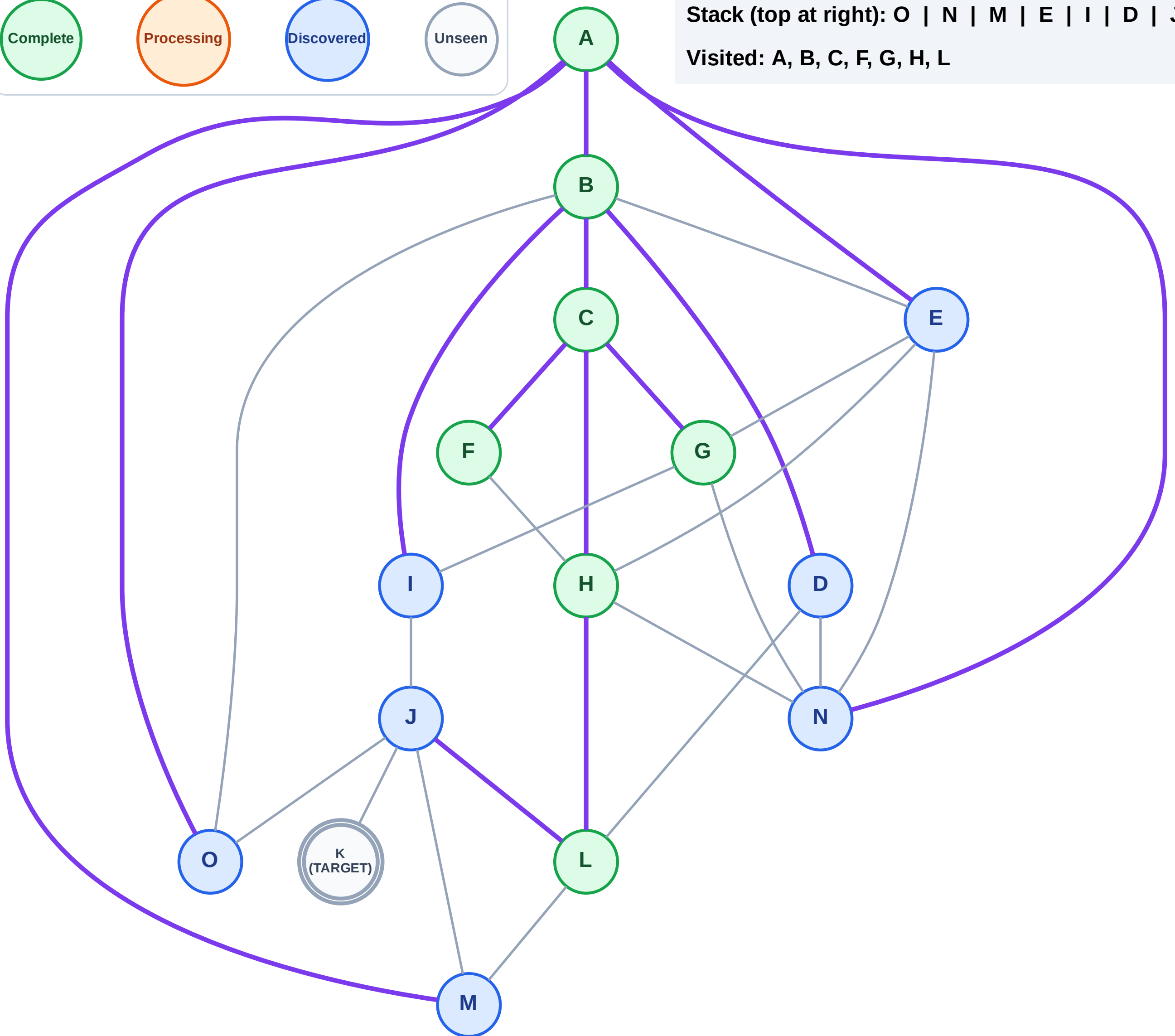
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D | J

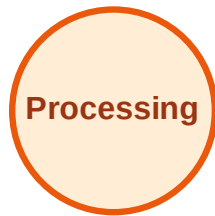
Visited: A, B, C, F, G, H, L



# DFS Traversal

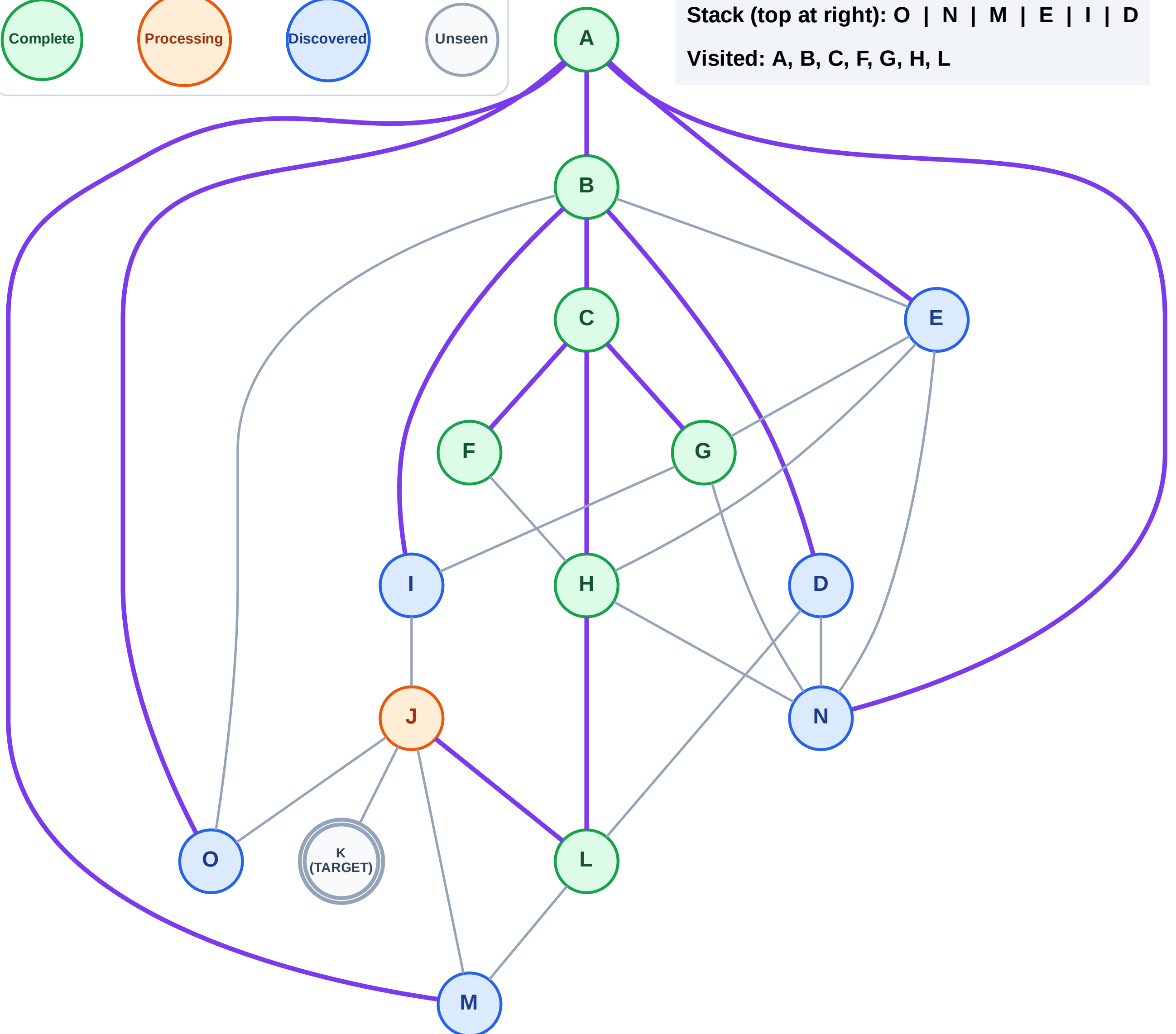
Step 16: Process node J

## Legend



Stack (top at right): O | N | M | E | I | D

Visited: A, B, C, F, G, H, L



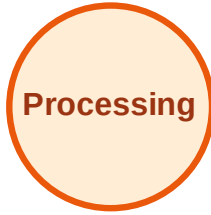
# DFS Traversal

Step 17: Discover K from J

## Legend



Complete



Processing



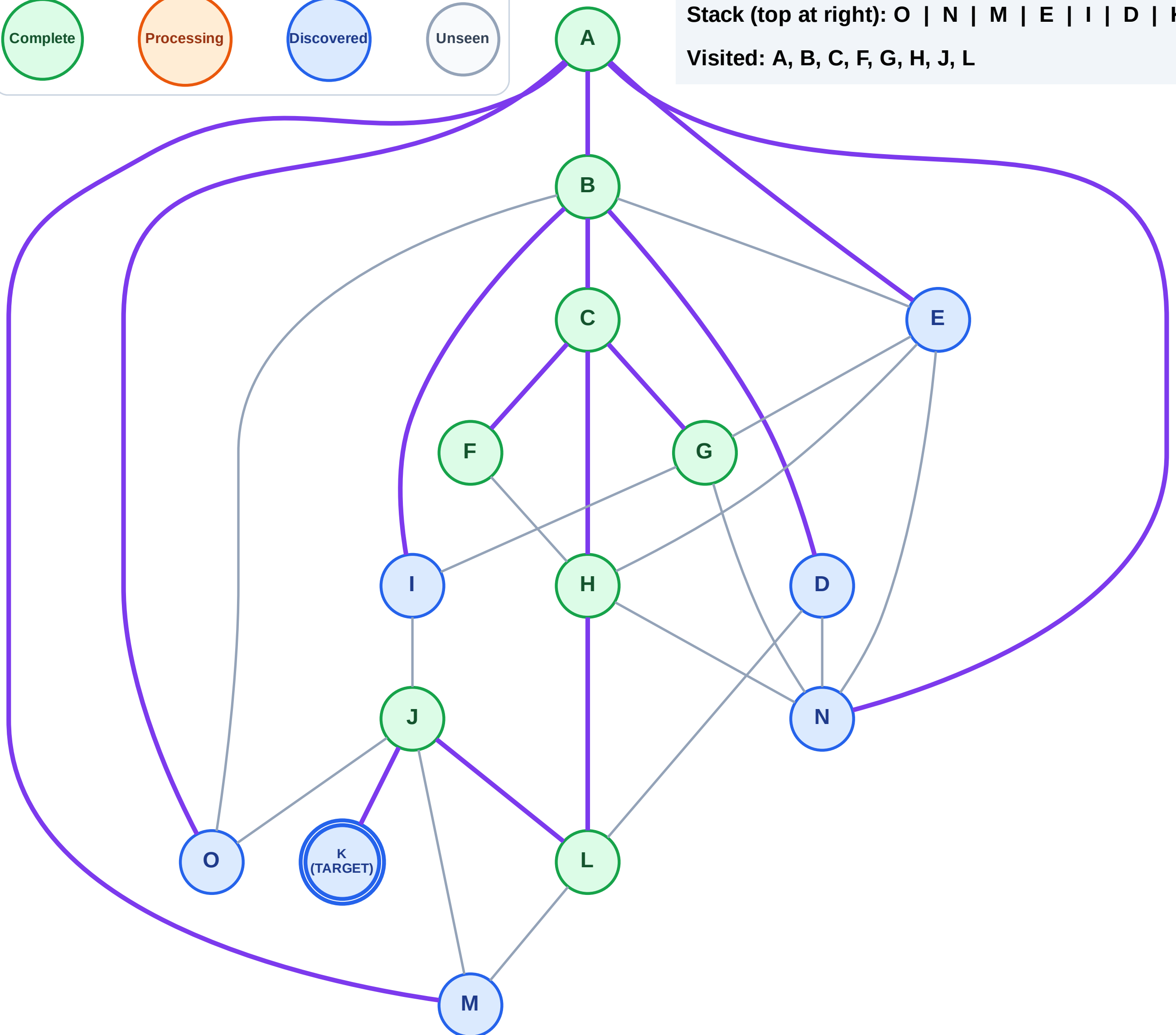
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D | K

Visited: A, B, C, F, G, H, J, L



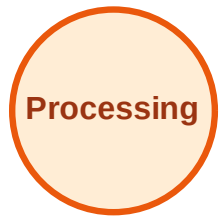
# DFS Traversal

Step 18: Process node K

## Legend



Complete



Processing



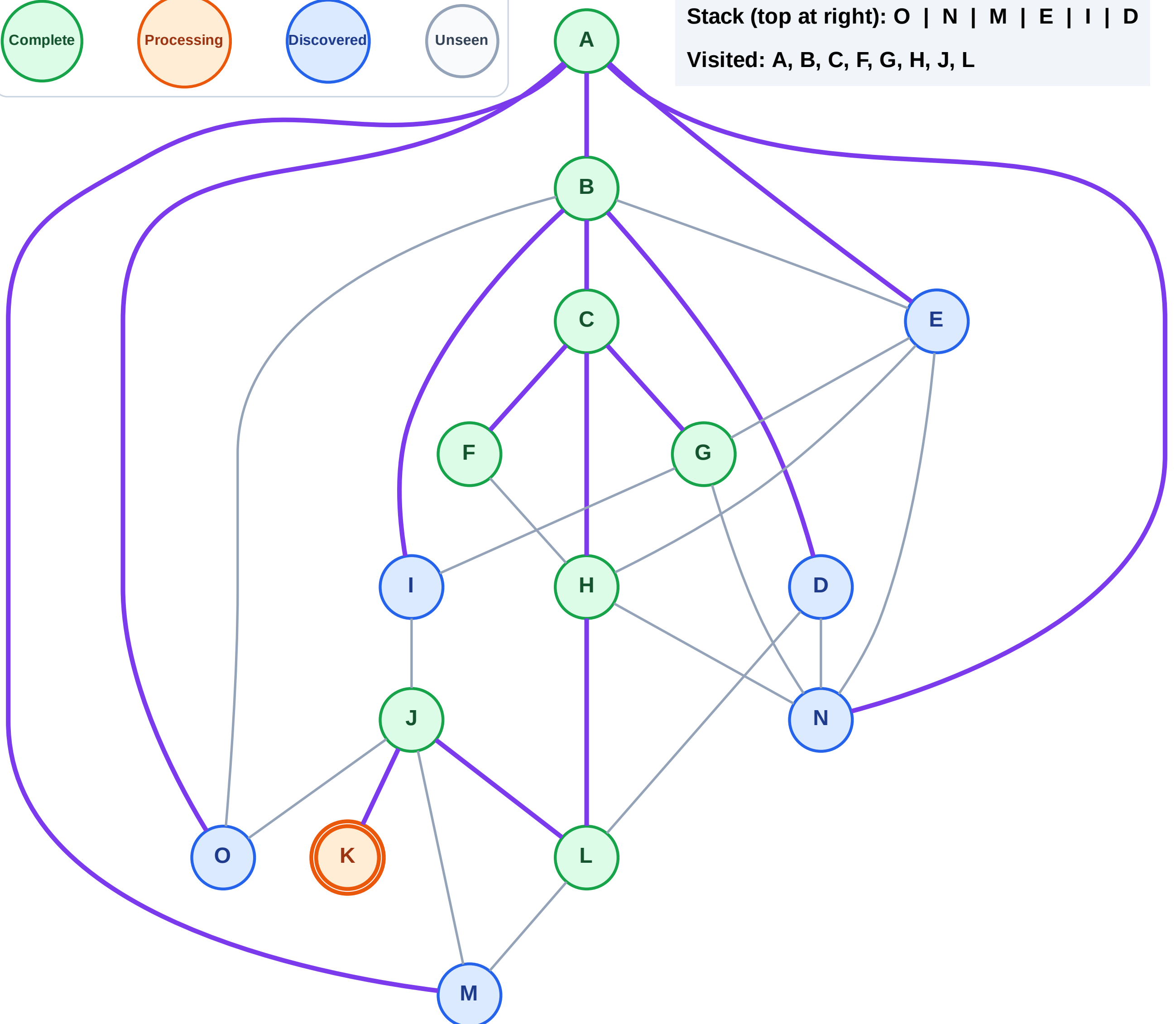
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D

Visited: A, B, C, F, G, H, J, L



# DFS Traversal

Step 19: TARGET NODE K REACHED! Path (6 hops): A → B → C → H → L → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | M | E | I | D  
Visited: A, B, C, F, G, H, J, K, L

